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(54) **CONTAINERS HAVING LEVER-BASED
ACTUATION OF BRAKES AND LOCKS**

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See application file for complete search history.

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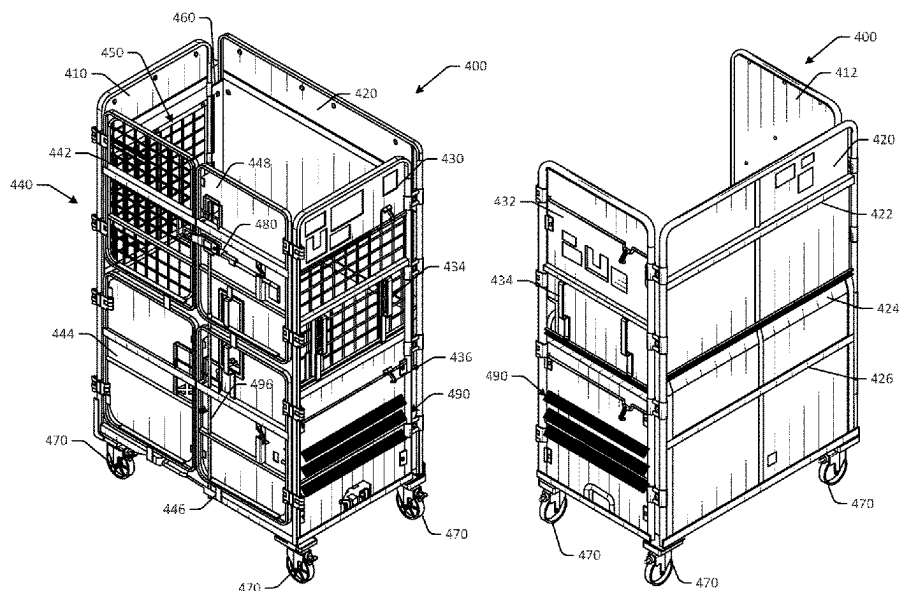
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(57) **ABSTRACT**

Systems and methods are disclosed for containers having lever-based actuation of brakes and locks. In one embodiment, an example container may include a first container wall, a second container wall, a third container wall, a fourth container wall, a first wheel, a second wheel, and a wheel lock assembly configured to prevent swiveling of the first wheel. The wheel lock assembly may include a first wheel lock, a first lever configured to actuate the first wheel lock, and a first cable that is coupled to the first wheel lock and the first lever, where the first lever is disposed on the third container wall, and a wheel brake assembly configured to prevent rotation of the second wheel, the wheel brake assembly including a first wheel brake, a second lever configured to actuate the first wheel brake, and a second cable coupled to the first wheel brake and the second lever.

20 Claims, 8 Drawing Sheets



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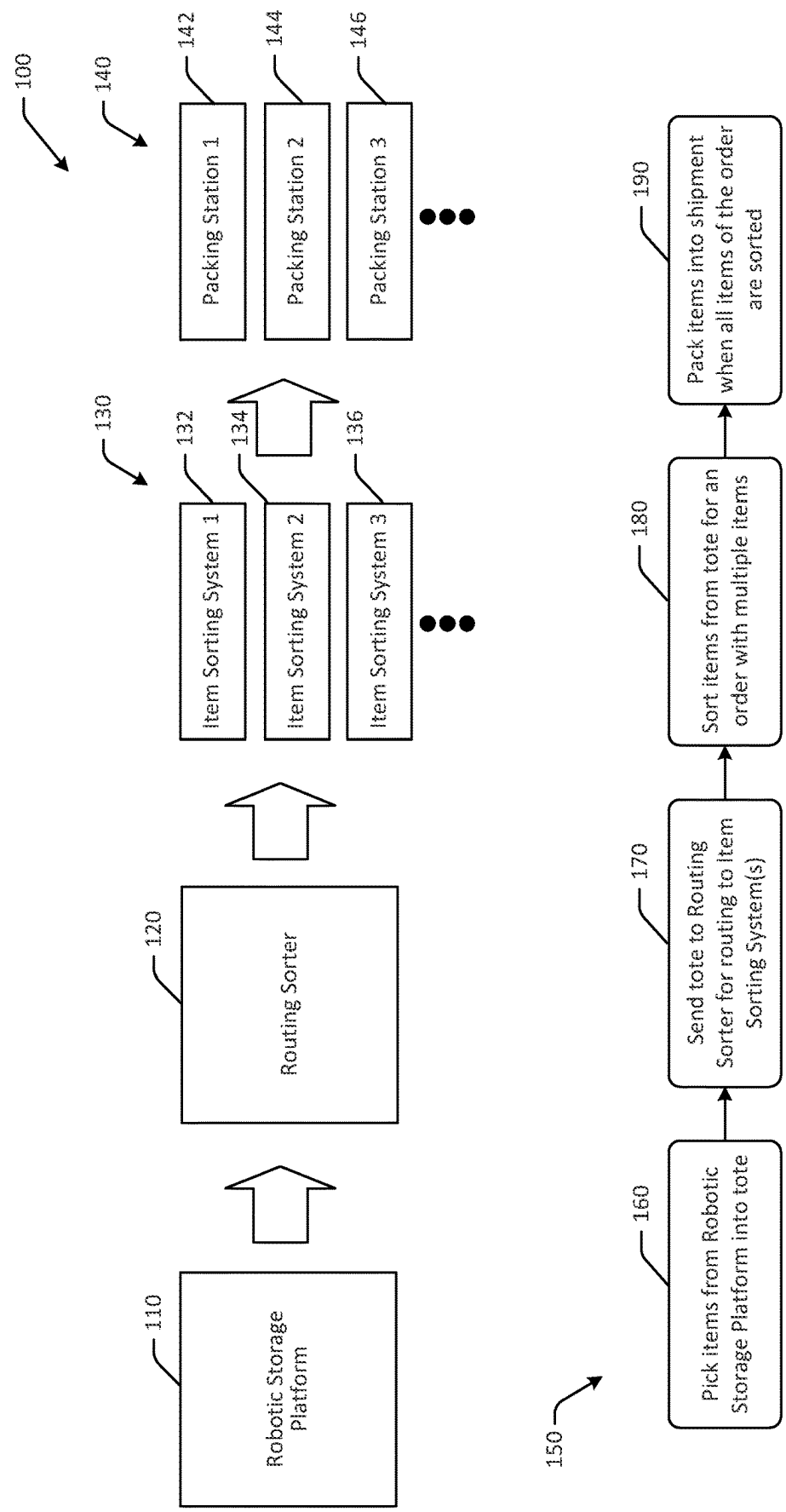


FIG. 1

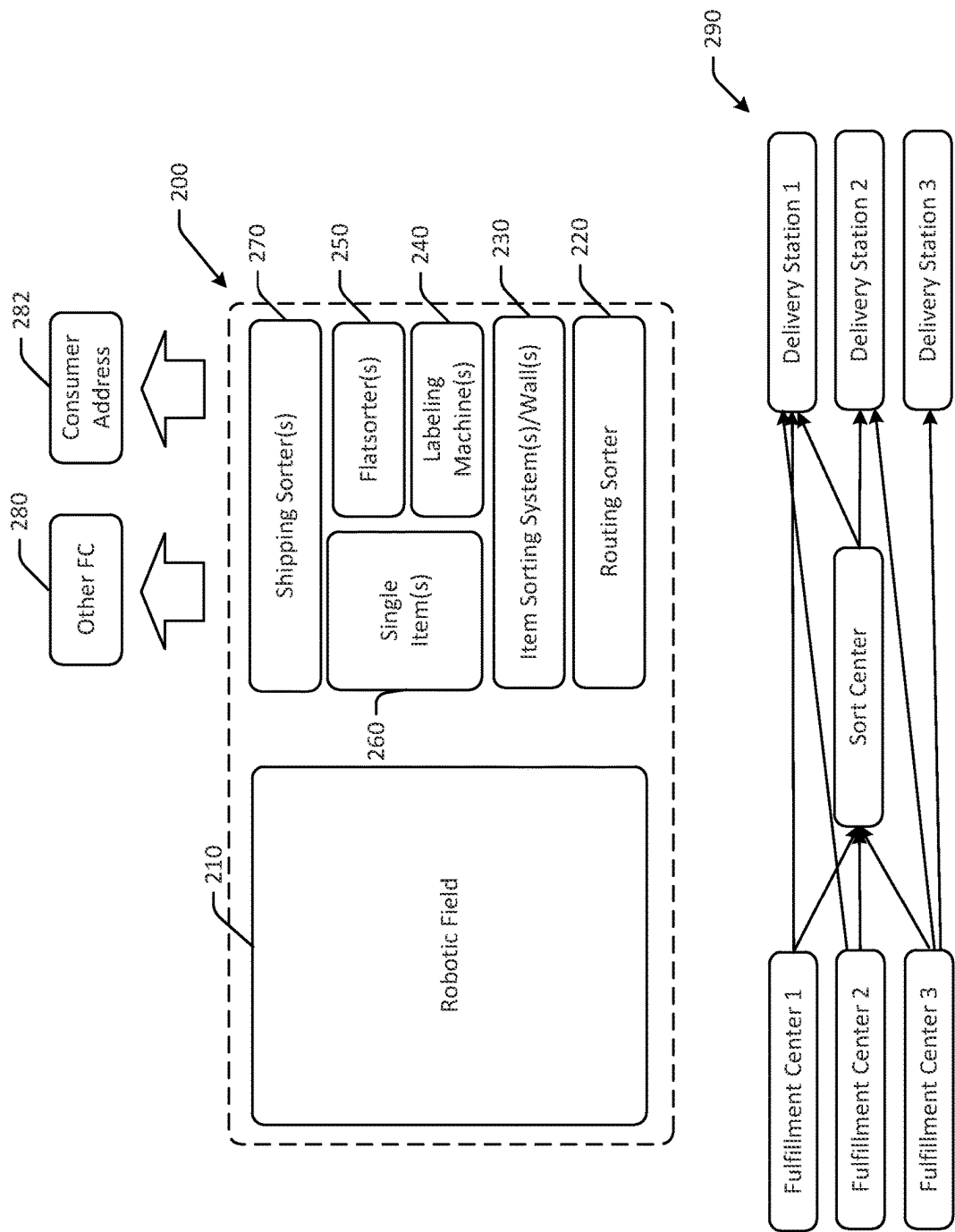


FIG. 2

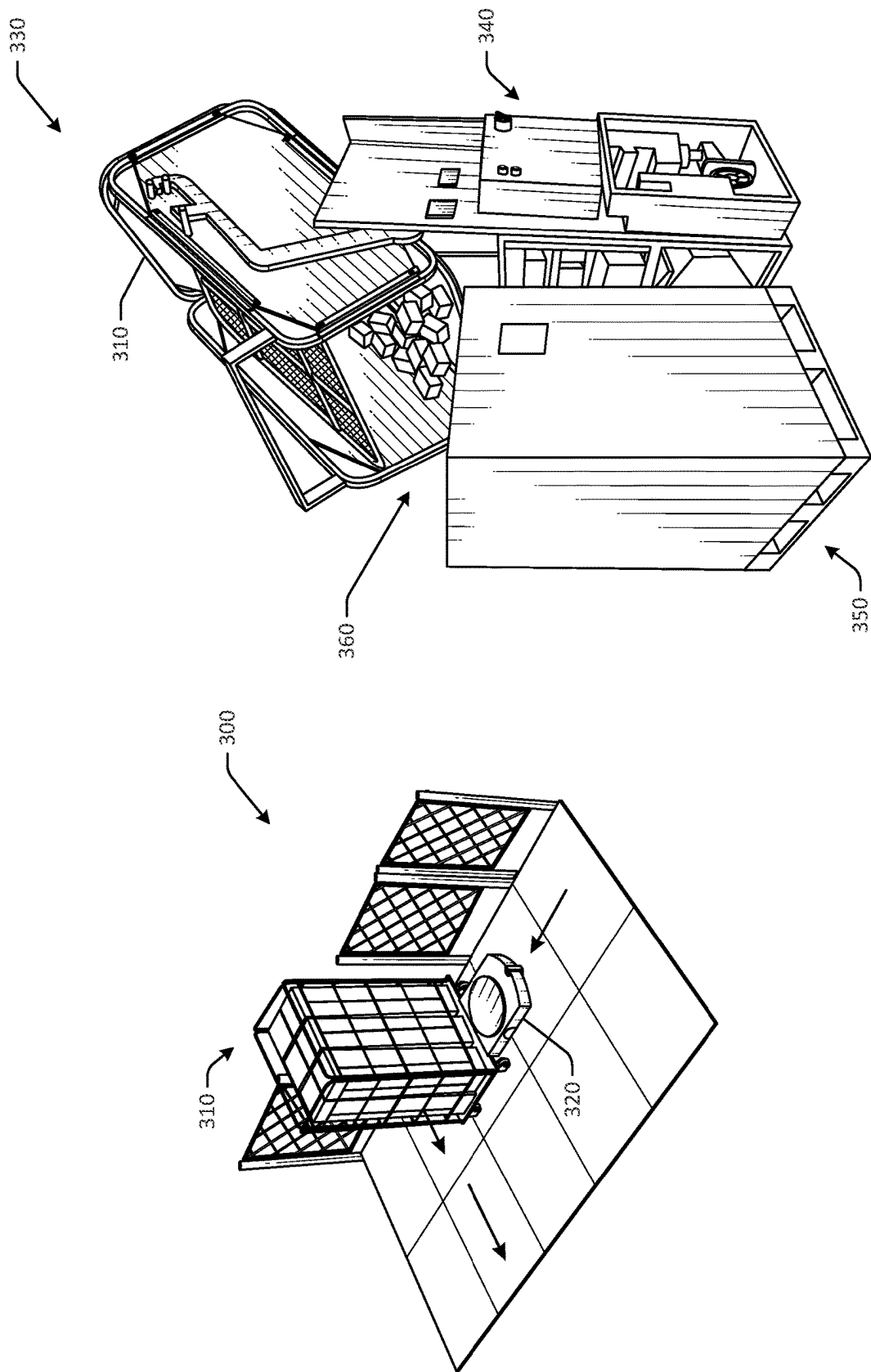


FIG. 3

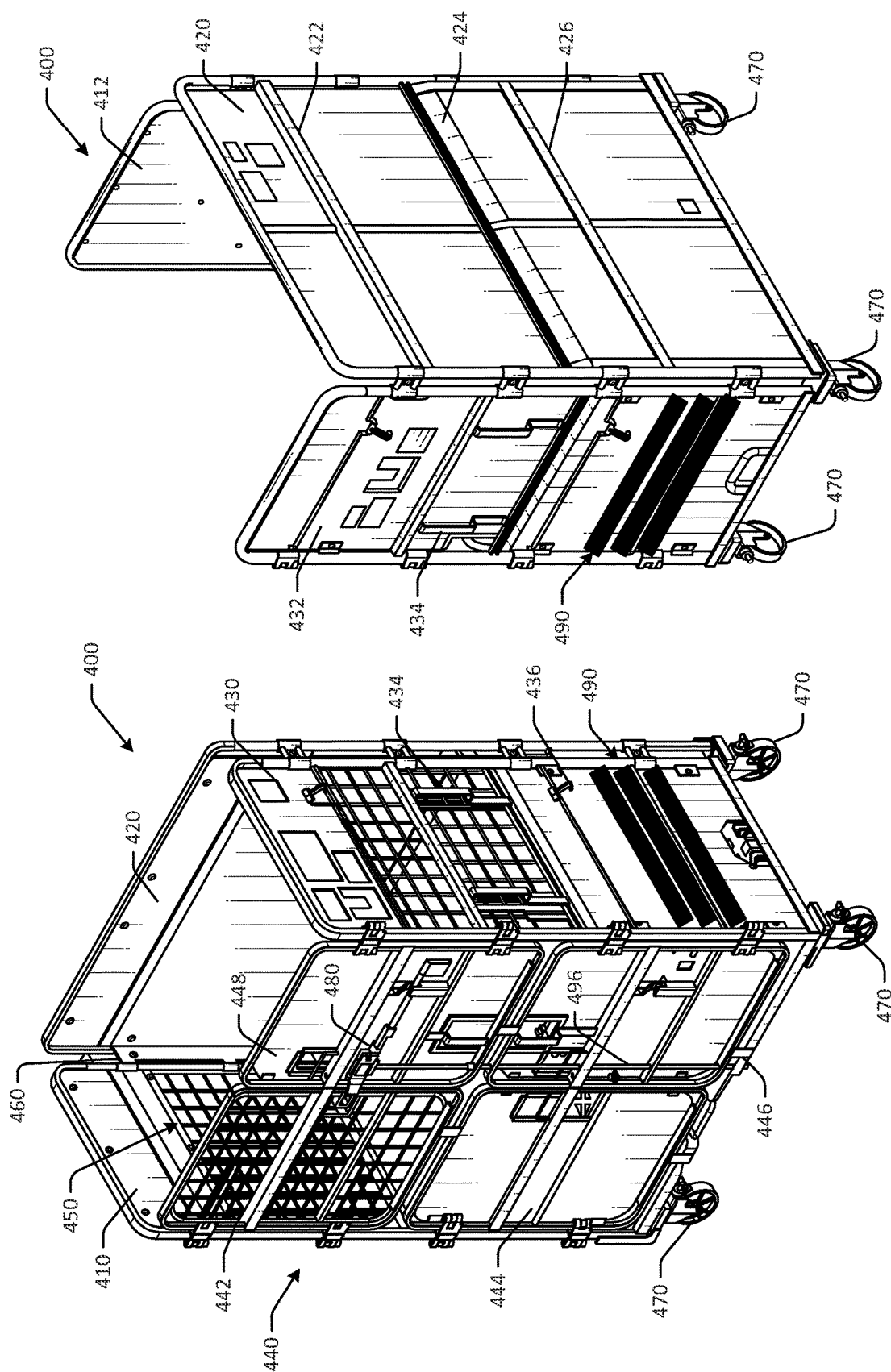


FIG. 4A

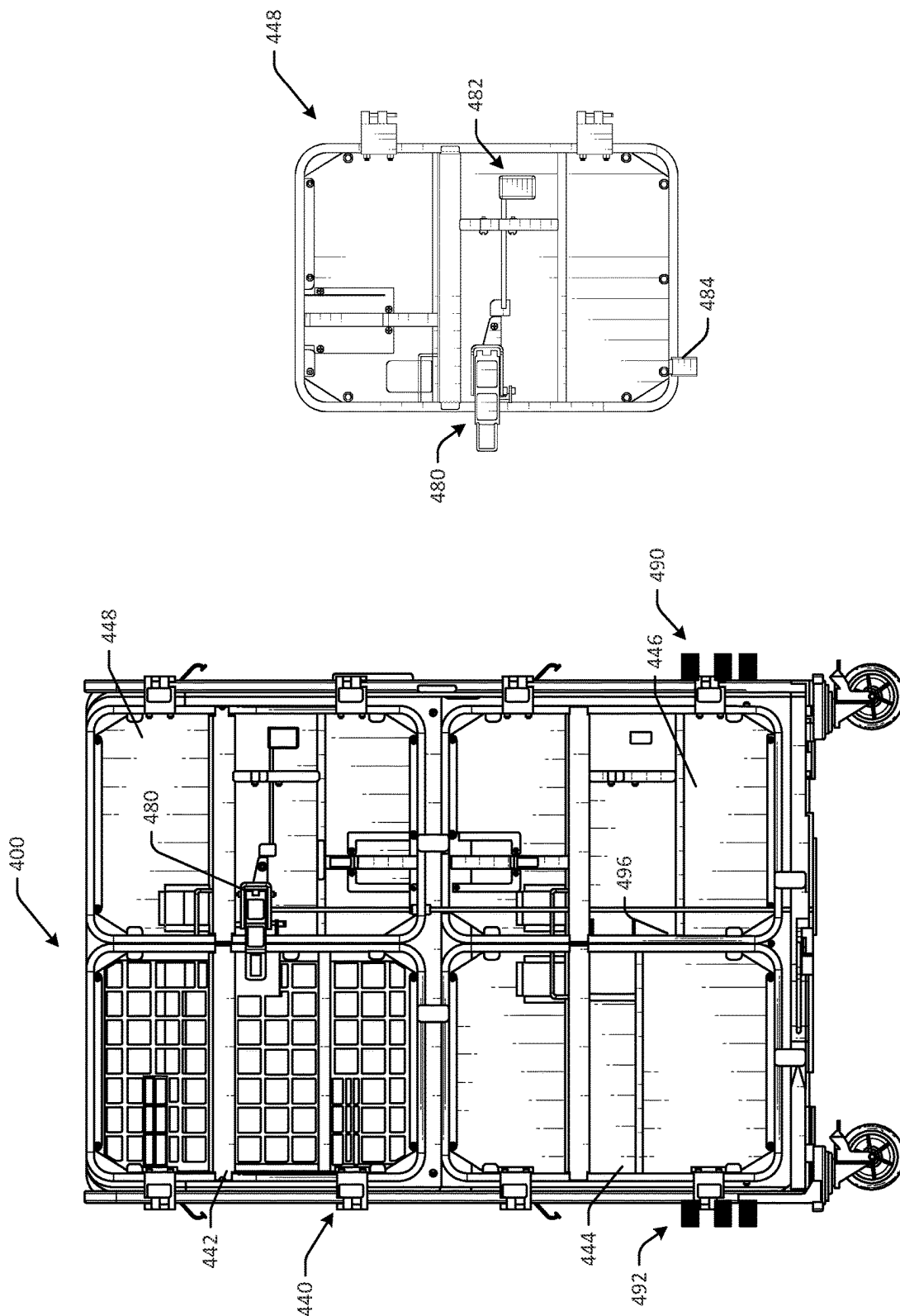


FIG. 4B

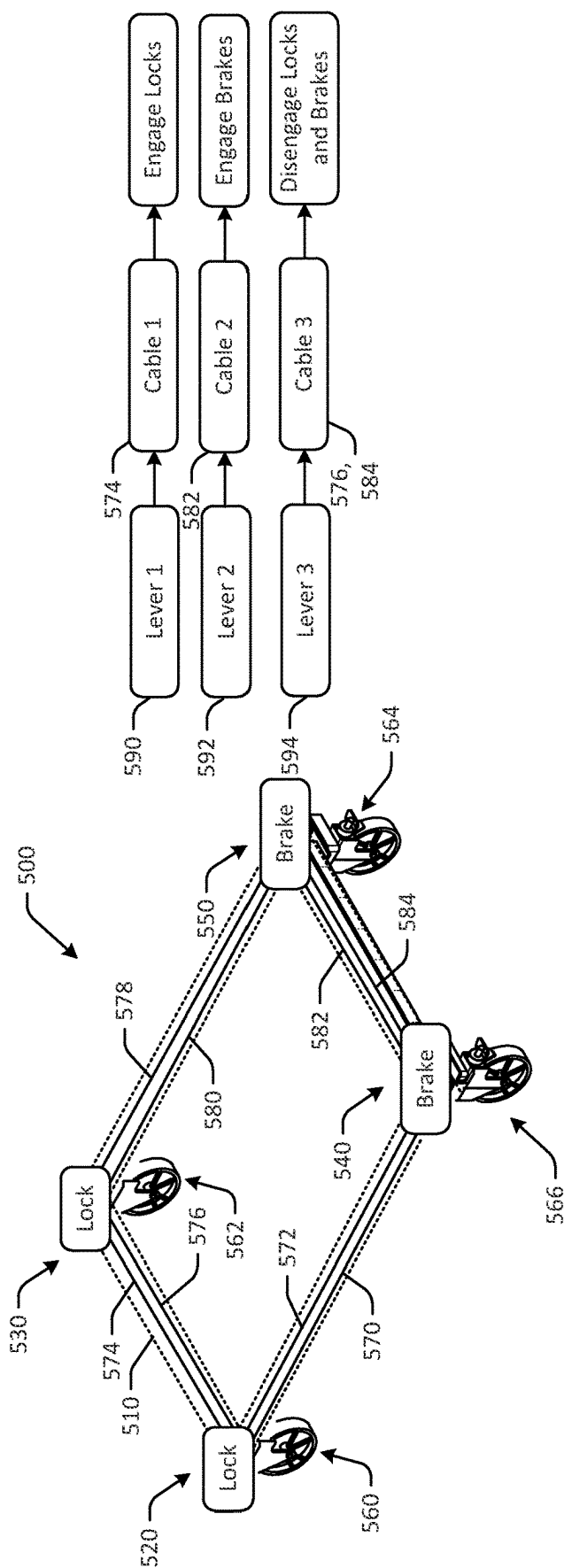


FIG. 5

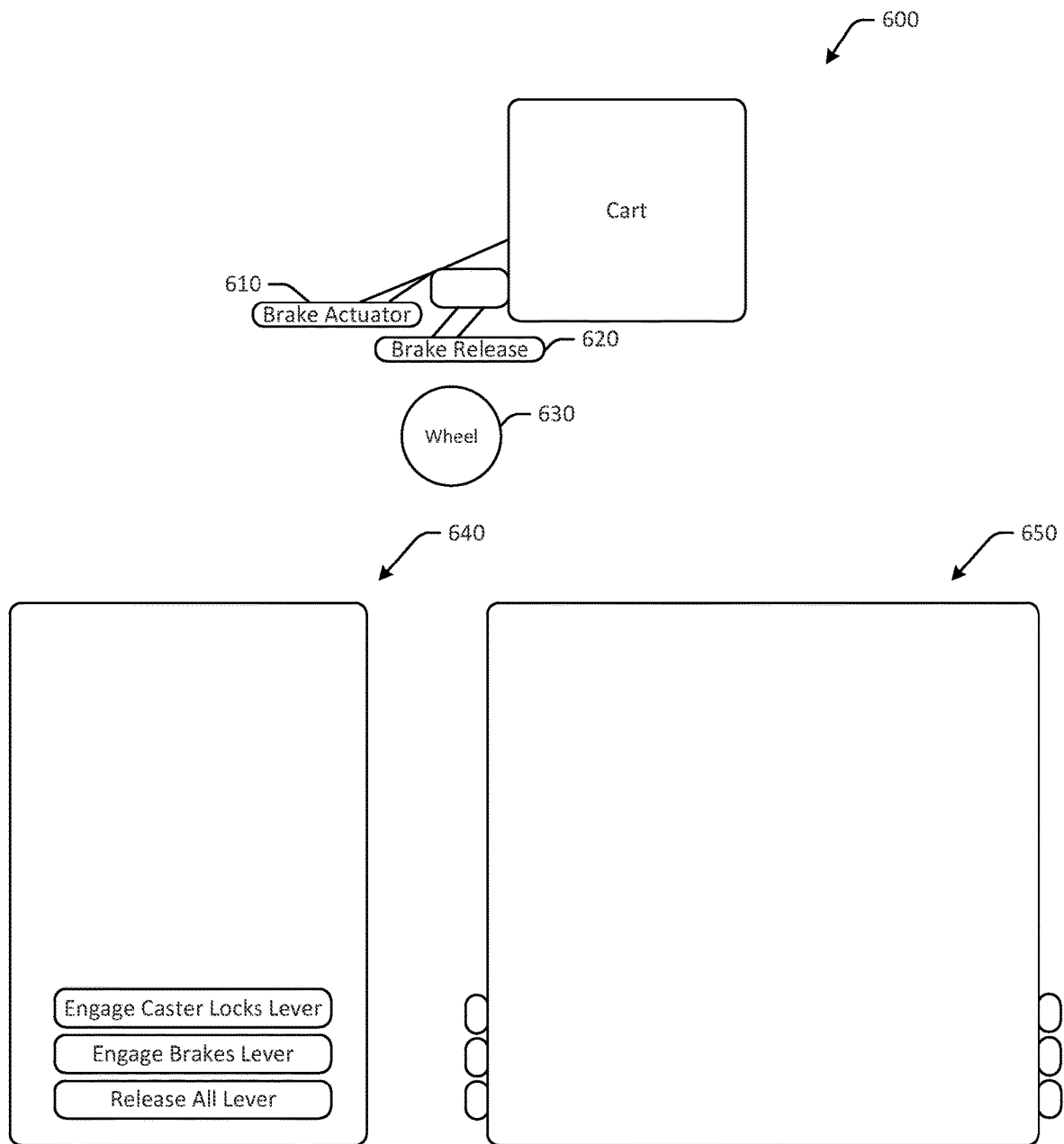


FIG. 6

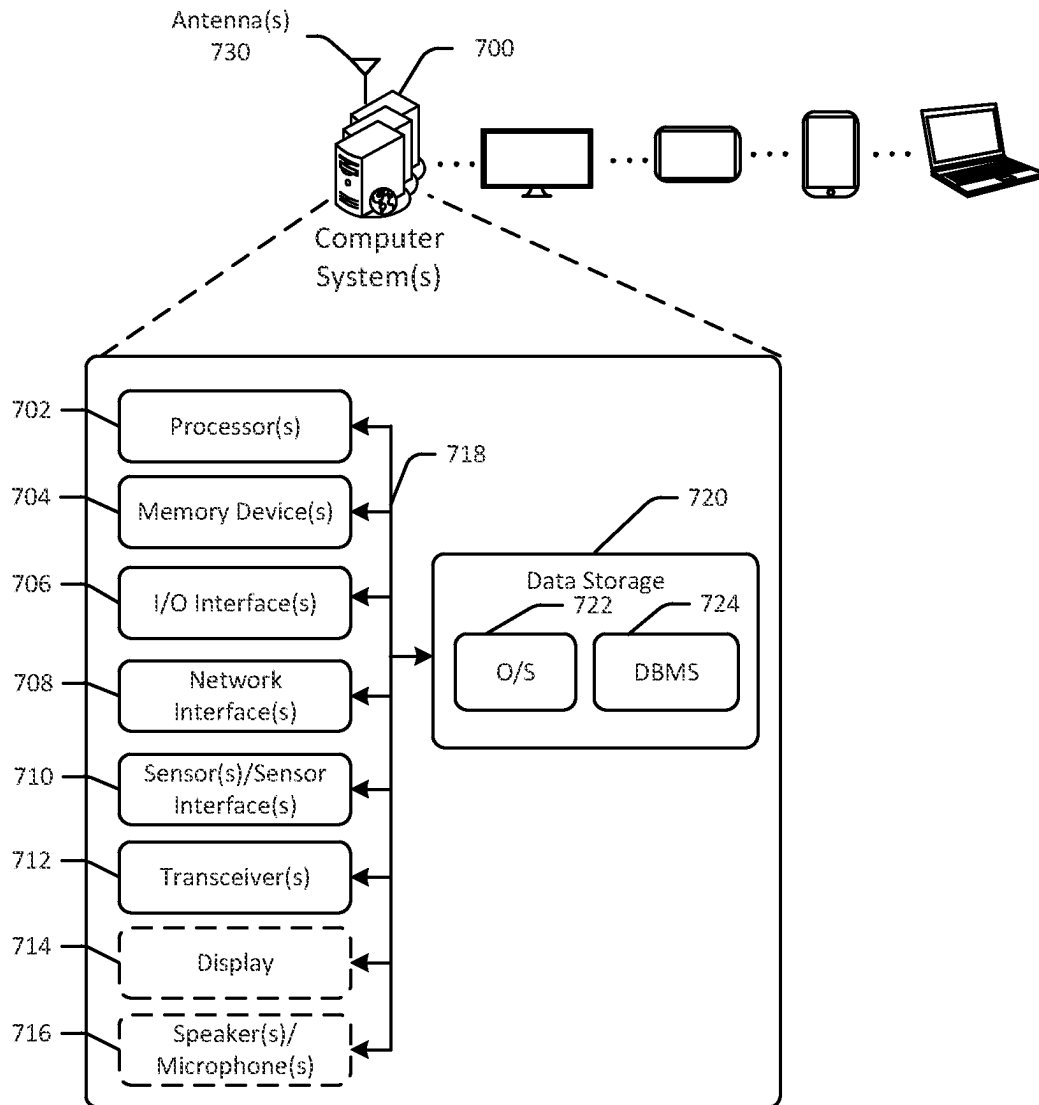


FIG. 7

CONTAINERS HAVING LEVER-BASED ACTUATION OF BRAKES AND LOCKS

BACKGROUND

As users increasingly make online purchases, fulfillment of such purchases and other orders may become increasingly complicated. For example, a fulfillment center may have output of upwards of one million packages per day. With such demands, efficiency of logistics related to processing orders and packages may be important. Accordingly, improvements in various operations of order fulfillment, such as improvements to picking technology, sorting technology, packing technology, and so forth may be desired, such that throughput can be increased and sustainability can be improved.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a hybrid schematic illustration of an example use case for containers having lever-based actuation of brakes and locks and an example process flow in accordance with one or more embodiments of the disclosure.

FIG. 2 is a hybrid schematic illustration of an example use case for containers having lever-based actuation of brakes and locks throughout a facility and between facilities in accordance with one or more embodiments of the disclosure.

FIG. 3 is a schematic illustration of a container being transported using an autonomous robot, and a container being dumped.

FIGS. 4A-4B are schematic illustrations of an example container having lever-based actuation of brakes and locks in accordance with one or more embodiments of the disclosure.

FIG. 5 is a schematic illustration of a container having lever-based actuation of brakes and locks in accordance with one or more embodiments of the disclosure.

FIG. 6 is a schematic illustration of various views of a container having lever-based actuation of brakes and locks in accordance with one or more embodiments of the disclosure.

FIG. 7 schematically illustrates an example architecture of a computer system associated with a robotic system in accordance with one or more embodiments of the disclosure.

The detailed description is set forth with reference to the accompanying drawings. The drawings are provided for purposes of illustration only and merely depict example embodiments of the disclosure. The drawings are provided to facilitate understanding of the disclosure and shall not be deemed to limit the breadth, scope, or applicability of the disclosure. The use of the same reference numerals indicates similar, but not necessarily the same or identical components. Different reference numerals may be used to identify similar components. Various embodiments may utilize elements or components other than those illustrated in the drawings, and some elements and/or components may not be present in various embodiments. The use of singular terminology to describe a component or element may, depending on the context, encompass a plural number of such components or elements and vice versa.

DETAILED DESCRIPTION

Overview

Fulfillment centers may be used to fulfill online purchases and other orders. For example, fulfillment centers may

include product inventory that may be pulled when an order for a particular product or multiple products is placed. In some instances, the product(s) may be packed and shipped from the fulfillment center. However, the process of obtaining the product(s), packing the product(s), and shipping the product(s) may be complicated due to the amount of inventory, the number of orders to process, the size of the fulfillment center, and/or other factors. In addition, a portion of the fulfillment center designated for packing or shipping may be different than the portion of the fulfillment center designated for holding product inventory. As a result, transportation of products and/or shipping of packages in an order may be time consuming.

In some instances, orders for products may include multiple items. For example, a user may place an order for two or more products. In such instances, the products that are ordered may not be in the same location of the fulfillment center, or one of the products may take a longer time to obtain or pick than the others. As a result, packing of the order may be delayed until all of the items in the order are ready for packing. To improve the speed of processing orders, in certain instances, robots and other technology may be deployed, such that manual efforts can be redirected to other tasks. For example, robots may be used to assist with locating products in an order during a pick process. However, directing picked products to the appropriate packing station and/or location may form a bottleneck in the operation of the fulfillment center. For example, after products are picked, the products may be placed in a container, such as a tote or other container, and directed to sortation machines to direct the picked products to the appropriate packing location. Products in the same order may be directed to the same packing location for consolidation and subsequent packing. However, a tote or other container may include products that are to be sorted to multiple different packing locations, and the sortation process may be slowed by sheer volume of products that are to be processed and/or sorted. In addition, in some instances, items that are part of the same order may not all be in the same fulfillment center or other location. For example, a first product in an order of two items may be at a first fulfillment center, and a second product in the same order may be at a second fulfillment center. In such instances, instead of shipping the two items in the order separately, such as one from each fulfillment center, items may be transferred from one fulfillment center to another, and then aggregated with other items in the order and shipped together. Such fulfillment center-to-fulfillment center transfers of items may be processed similar to items that are to be shipped to consumers. For example, the items to be transferred may be picked, routed to a sortation machine, sorted into a particular container (e.g., a container designated for a particular fulfillment center, etc.), packed, and sent. In some instances, containers destined for other fulfillment centers may be infinite bottom containers, or containers that may be filled without consideration of a level of fullness or remaining capacity of the container. In such instances, capacity of the containers may be monitored externally (e.g., by a user manually, by a camera system automatically, using different sensors, etc.). Full containers may be removed from a sortation system and replaced with an empty container to continue aggregating items destined for a fulfillment center.

As items are placed into containers or packages for transport, sortation, etc., the packages may be placed in foldable or non-foldable containers that are configured to hold a plurality of packages. For example, a foldable container may be filled with packages that are destined for

another fulfillment center or other facility. The foldable containers described herein may be used not only to store and transport packages or other items within a facility, but may also be used to transport the packages to another facility. For example, in some embodiments, the foldable containers may be filled with packages, and the filled foldable container may be loaded onto a truck and transported to another facility. At the destination facility, the entire foldable container may be unloaded from the truck and scanned to identify the contents of the foldable container, without having to empty the foldable container. This may allow for increased efficiency and reduced handling of individual packages, and may also provide the ability for foldable containers to be used across any number of facilities and/or workstations within a facility.

Once the foldable container is unloaded from a truck, or otherwise reaches its destination within a facility (e.g., if the foldable container does not leave a facility, etc.), the foldable container may be dumped so as to remove its contents (e.g., from its open top, etc.). For example, the foldable container may be dumped onto a conveyor, such that the packages or other items inside the foldable container fall onto the conveyor. Foldable containers may be dumped manually (e.g., using equipment, etc.) or automatically.

When carts or other foldable containers are transported manually, operators may be aware of the orientation of the cart and may position the cart such that certain features, such as wheel locks, are easily accessible. However, when robots transport such carts, the robots may not be aware of the particular orientation of a cart. In one example, when loading carts into a trailer, a robot may not be aware of the location of cart brake releases and may load the cart or other container in a suboptimal orientation.

Embodiments of the disclosure include foldable or non-foldable containers having lever-based actuation of brakes and locks. Some embodiments include actuators that are accessible from any side and/or corner of the container or cart, so as to allow for transport and positioning of the container without a need to consider the orientation of the container. Containers may include caster wheels having caster locks (that prevent swiveling of certain wheels) and brakes (that prevent rotation or rolling of certain wheels). Embodiments may include controls for actuation of the caster locks and brakes that can be actuated by the foot or hand of an operator that allow for caster locking and/or braking of the cart from any side or corner of the container. Embodiments may provide an ergonomic and convenient actuation mechanism that allows for the containers to physically fold up to minimize storage space when not being used and to enable efficient reverse logistics.

Embodiments of the disclosure therefore include foldable containers having lever-based actuation of brakes and locks that maintain overall container volume and improve ergonomics, improve ejection rates of packages or other objects, reduce processing time, and increase throughput by avoiding jams. Handling of the foldable container within and/or between facilities may be manual (e.g., by an operator), automated (e.g., by an autonomous robot or other vehicle), and/or a combination thereof. Accordingly, the configuration of foldable containers described herein may be compatible with autonomous robots, as well as with manual manipulation, while at the same time providing rigid support for transport and/or storage of items or packages within and between facilities. Embodiments may provide ergonomic features to facilitate manual handling of foldable containers. U.S. patent application Ser. No. 17/545,356, titled "Foldable Containers with Improved Dumping Features," describing

various aspects of a foldable container, and U.S. patent application Ser. No. 17/202,693, titled "Foldable Containers for Manual and Robotic Use," describing various features of a foldable container for manual and robotic use, are both hereby incorporated by reference in their entireties.

The embodiments described herein may improve the functionality of carts or containers deployed in a facility. For example, work cells where packages or items are loaded into containers may include bays into which containers are pushed. Unlike current containers with actuators on each corner of the container (where each individual wheel is independently actuated), embodiments may be pushed into the bays in front-facing, side-facing, or rear-facing orientations, as the caster locks and brakes can be actuated from either side of the container, thereby allowing increased process flexibility and avoiding a need to program or develop technology for autonomous robots to determine container orientation. In contrast to existing containers, containers as described herein are not sensitive to orientation during placement inside a facility, truck, or other location, which may increase process flexibility by avoiding a need for positive determination of orientation, and avoid restrictions on orientation or placement of the container at different sites in a facility. As a result, efficiency and robustness of automated processes involving such containers is increased. Certain embodiments may be configured to allow for various lock and/or brake engagement states, such as brakes engaged (e.g., cart not allowed to move, etc.), caster locks engaged and brakes disengaged (e.g., cart being moved from one location to another, as caster locks make the cart stable while being moved, etc.), caster locks and brakes both engaged (e.g., no motion is desired and rotational stability is desired, etc.).

Embodiments may therefore provide a number of benefits, including allowing control of the braking wheels via accessible actuators from either short side of the cart, including the corners, without adding any thickness to the short side of the cart. A single motion can be used to set multiple wheels, for example, releasing all brakes/locks (maneuvering mode), setting all brakes (stationary mode) and setting caster locks only.

Foldable containers may be folded into a folded L-shaped configuration to conserve space when not in use, and may be unfolded into a rectangular configuration when in use. Embodiments may be configured for both manual and robotic use. For example, foldable carts as described herein may be configured to be pushed, pulled, or otherwise manipulated manually by a human operator, and may also be lifted and transported by a robot. This flexibility may provide improved dual usability for various functions within a fulfillment center, and may reduce or eliminate the need for different container designs for manual and robotic use. In addition, the foldable containers described herein may be configured to provide ergonomic support for manual use, while also providing structural rigidity to be lifted from a bottom surface using a robot.

In some embodiments, foldable containers as described herein may be loaded with packages destined for a delivery station. The foldable containers may be sorted and transported while the packages or other items remain in the foldable containers. Sortation and/or movement of the foldable containers may be performed using robots and/or manually.

Embodiments of the disclosure include foldable containers having lever-based actuation of brakes and locks that can be moved by both robots and operators. For example, foldable containers may be pushed, pulled, or otherwise

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manipulated manually by human operators, and may also be lifted and moved or transported by robots. Some robots may lift the foldable containers from a bottom surface or a bottom platform. The foldable containers may be configured to fold and be stored or stacked in a nested configuration to save space as well.

Embodiments may therefore increase throughput and speed of consolidating items for multi-item orders and/or consolidating packages that are destined for certain related destinations, such as other fulfillment centers. Some embodiments include optimized process flows for processing of orders at fulfillment centers, as well as process flows or equipment to increase speed of consolidating products in a multi-item order and/or speed of sorting packages. As a result, throughput of fulfillment centers may be improved, and/or logistics of fulfillment center operations may be less complicated.

Referring to FIG. 1, an example use case **100** for containers having lever-based actuation of brakes and locks and an example process flow is depicted in accordance with one or more embodiments of the disclosure. Although discussed in the context of online orders, other embodiments may be directed to any suitable use case where products are picked and sorted, or packages are sorted, such as instances where users may pick up orders rather than receiving a shipment, instances where items are aggregated for transport to another fulfillment center, and so forth.

In FIG. 1, a fulfillment center may include a robotic storage platform **110**, a routing sorter **120**, one or more item sorting systems **130**, and one or more packing stations **140**. The robotic storage platform **110** may be a portion of the fulfillment center at which products picked from product inventory are placed. Inventory may be stored in containers in foldable containers in some instances. Robots may be used to pick products from inventory and to deliver to the robotic storage platform in some instances, while in other instances, manual effort or a combination thereof may be used to pick products. The picking process at the robotic storage platform may include locating a product in an order, obtaining the product, and sending the product to the robotic storage platform **110**, such as via a conveyor belt. In the illustrated embodiment, products at the robotic storage platform **110** may be placed in a container, such as a tote. The tote may be assigned to, or otherwise associated with, a particular item sorting system machine in some instances. For example, a certain tote may be associated with a certain item sorting system, such that products that are designated to be picked and placed in the tote are for orders that are to be consolidated at that particular item sorting system. The association between the tote and the item sorting system may be static in some instances. In other embodiments, there may not be any association between totes and item sorting systems, or associations may be dynamic.

At the routing sorter **120**, totes including products that have been picked may be routed to the appropriate or designated item sorting system. For example, the routing sorter **120** may optionally determine an identifier associated with the tote, and may determine one or more item sorting systems to which the tote is to be routed using the identifier or using another factor, such as sortation system load. The routing sorter **120** may route or direct the tote to an item sorting system.

The item sorting systems **130** may include one or more item sorting system machines. In FIG. 1, a first item sorting system **132**, a second item sorting system **134**, a third item sorting system **136**, and so forth may be included. Any number of item sorting systems may be included. Some or

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all of the item sorting systems may optionally be associated with certain totes. The item sorting systems may be used to consolidate or otherwise aggregate products for single or multi-item orders and/or for transfer to a different fulfillment center. For example, a first tote may include a first item of a multi-item order, and a second tote may include a second item of the multi-item order. The item sorting system may therefore identify the orders associated with the respective products in a tote, and may transport the products to a container, such as a tote, a flexible container, a specific chute leading to a container, or a different container associated with the order. When the order is complete with all of the products in the associated chute or container, the order may be packed. In instances where a container is designated for a different fulfillment center, as opposed to an online order, the container may be packed when full, as opposed to when certain items are placed into the container (e.g., there may not be any specific items that need to be in the container before packing, rather, the container may just be a certain threshold full, etc.). Accordingly, a specific item sorting system may be designated for fulfillment of a particular multi-item order. As a result, all of the products in the multi-item order may be placed in totes that are directed to that particular item sorting system. At the item sorting systems **130**, totes that are received via the routing sorter **120** may be emptied, and the products in the respective totes may be transported to the appropriate chutes or containers for the orders for which the products were picked.

After a single or multi-item order is complete (e.g., the item sorting system has delivered all of the products in the order to the appropriate chute, container, etc.), or when a container designated for another fulfillment center is full (where full is a configurable threshold, such as about 60% full capacity, 70% full capacity, 80% full capacity, 90% full capacity, etc.), the order may be packed at the packing station **140**. In some embodiments, one or more packing stations may be included. In some instances, a packing station may service more than one item sorting system, while in other instances, more than one packing station may service one item sorting system. In the illustration of FIG. 1, a first packing station **142** may be used to pack orders from the first item sorting system **132**, a second packing station **144** may be used to pack orders from the second item sorting system **134**, a third packing station **146** may be used to pack orders from the third item sorting system **136**, and so forth. At the packing stations **140**, the orders may be placed into boxes and sealed for subsequent shipment. The packages may then be processed for shipment to the user. For example, the packages may be placed into foldable containers for shipment to another fulfillment center, a sort center, or another facility. At the subsequent facility, the foldable container may be dumped to remove its contents.

At the fulfillment center, an example process flow **150** illustrated in FIG. 1 may be implemented to improve the efficiency and/or throughput of the fulfillment center. At a first block **160**, items may be picked from the robotic storage platform **110** into a tote that may optionally be associated with a specific item sorting system. At a second block **170**, the tote may be sent to the routing sorter **120** for routing to an item sorting system. At a third block **180**, the items from the tote may be sorted for an order with multiple item by the specific item sorting system. At a fourth block **190**, the items may be packed into a shipment when all of the items in the order are sorted.

FIG. 2 is a hybrid schematic illustration of an example use case for containers having lever-based actuation of brakes and locks throughout a facility and between facilities in

accordance with one or more embodiments of the disclosure. Other embodiments may include additional or fewer components.

In FIG. 2, an example layout of a fulfillment center **200** is depicted. The fulfillment center **200** may include a robotic field **210** at which product inventory may be stored for picking (e.g., optionally in one or more flexible container pods, etc.), one or more routing sorters **220** that may be used to direct totes or other containers to item sorting systems, one or more item sorting systems or walls **230** used to consolidate products for multi-item orders and/or to pack multi-item orders, one or more single item sections **260** that may be used to pack single item orders, one or more labeling machines **240** that may be used to apply shipping labels to packages, one or more flat sorters **250** and shipping sorters **270** to sort labeled shipments (e.g., by destination, carrier, etc.) for pickup from the fulfillment center **200**.

In some embodiments, the item sorting systems described herein may be a part of the flat sorters **250**, where the item sorting systems may be configured to sort packages into containers or chutes. In such embodiments, the item sorting systems may or may not also be used at the item sorting systems **230** portion of the fulfillment center **200**. Accordingly, the item sorting systems may be disposed at, or otherwise coupled to, a cross belt conveyor system, such as the flat sorters **250** of the fulfillment center **200**.

The item sorting system machines **230** may include containers and/or containers of different sizes (e.g., small, medium, large, etc.) and may be configured, in one example, to handle items that weigh up to twenty or more pounds (e.g., 100 pounds or more, etc.). In some embodiments, the item sorting system machines **230** may include multiple chutes, such as about 328 chutes, and may be configured to sort items at a rate of about 2,100 units per hour or more. In some instances, the item sorting system machines **230** may have two inductors (e.g., one on each side, etc.), and may be modular. For example, the item sorting system machines **230** may each include sixteen expansion modules, where expansion modules may be defined as three two-sided columns next to one another for a total length of about 80 feet. The item sorting system machines **230** may reduce labor and capital costs associated with processing orders.

In some embodiments, the item sorting system **230** may replace other processes, such as manual processes. The item sorting system **230** may be a cross-belt shuttle sorter that sorts singulated products into containers or totes. Item sorting systems **230** may be capable of sorting at a rate of 2,100 units per hour or more. Certain item sorting systems **230** may be configured to handle items of up to twenty pounds, or more in some instances, with dimensions of about 18"×14"×8" or greater, which may cover almost all products at the fulfillment center **200**. The item sorting systems **230** may operate as a high-speed, high-destination sort solution that intakes items or packages and sorts them into containers using a shuttle that travels vertically and horizontally inside the machine (or outside in some instances).

Individual item sorting system machines may be item sorting systems, and may include a number of, such as two or more, modular sorting machines coupled in series, or otherwise adjacent to each other and connected. The modular sorting machines may include a first modular sorting machine. The modular sorting machines may be configured to singulate items from a tote including a plurality of items into a plurality of chutes or containers (e.g. induct individual items from a container that has multiple items, and place the inducted items into the appropriate chute to be routed to a

container, where chutes or containers are associated with multi-item orders). The tote from which items are inducted may be associated with the individual item sorting system machine (e.g., the modular sorting machines that form the individual item sorting system machine, etc.). In some embodiments, item sorting systems or individual item sorting machines may be configured to induct and sort packages based at least in part on a destination of the respective packages. Destinations may be internal destinations within a fulfillment center, external destinations to geographic regions or addresses, or other destination types. For example, output from the fulfillment center **200** may include containers of items routed to other fulfillment centers **280**, packages addressed to consumer addresses **282**, and so forth.

Accordingly, in some embodiments, item sorting systems may be arranged in rows and may receive totes from a routing sorter, thereby streamlining fulfillment center operation and reducing labor and space costs. The item sorting systems may process totes for multi-order sortation and consolidation. As a result, there may no longer be a need to singulate and send items to a wall for manual extraction, because each tote may be assigned to a particular item sorting system machine. Induct stations can be replaced with item sorting system machines.

In another embodiment, pickers may pick items directly to a segmented belt conveyor at a station that may be near an item sorting system machine. Other nearby pick stations may also pick items directly to conveyance for the same item sorting system machine. Picked items being transported to a single item sorting system machine may merge together to be inducted into their unique item sorting system machine, where multi-item orders may be consolidated and sent to packing.

Some embodiments may be used across facilities in a fulfillment network. An example fulfillment network **290** is depicted in FIG. 2. The fulfillment network **290** may include a number of fulfillment centers, a number of sort centers, a number of delivery stations, and so forth. The fulfillment centers may be used to pack items for fulfillment. Packed items may be routed to a sort center for redirection to an appropriate delivery station. In some instances, packed items may be routed directly to a delivery station while bypassing sort centers. Embodiments of the foldable containers described herein may be configured to move from a first facility to a second facility, a third facility, and so on, so as to seamlessly move amongst different facilities in a fulfillment network. For example, a foldable container may move from a fulfillment center to a sort center, and then on to a delivery station or other facility, thereby providing a flexible and dynamic solution without having to unload contents of the foldable container. At any of the locations, the foldable containers may be dumped with reduced jams.

Embodiments of the disclosure include foldable containers having lever-based actuation of brakes and locks. The foldable containers may be used to store, at least temporarily, packages or other items. Certain embodiments may improve processing speed and/or throughput of fulfillment centers. Certain embodiments may improve performance of mechanical equipment for sortation and/or consolidation of items for multi-item orders via increased tolerances. While described in the context of online orders, aspects of this disclosure are more broadly applicable to other forms of product sortation.

Example embodiments of the disclosure provide a number of technical features or technical effects. For example, in accordance with example embodiments of the disclosure, certain embodiments of the disclosure may improve pro-

cessing speed, throughput, and/or efficiency of fulfillment centers. The above examples of technical features and/or technical effects of example embodiments of the disclosure are merely illustrative and not exhaustive.

One or more illustrative embodiments of the disclosure have been described above. The above-described embodiments are merely illustrative of the scope of this disclosure and are not intended to be limiting in any way. Accordingly, variations, modifications, and equivalents of the embodiments disclosed herein are also within the scope of this disclosure. The above-described embodiments and additional and/or alternative embodiments of the disclosure will be described in detail hereinafter through reference to the accompanying drawings.

ILLUSTRATIVE EMBODIMENTS AND USE CASES

FIG. 3 is a schematic illustration of a container being transported using an autonomous robot, and a container being dumped. Other embodiments may include additional or fewer components. The illustration of FIG. 3 is not to scale, and may not be illustrated to scale with respect to other figures.

In the example of FIG. 3, a foldable container 310 may be positioned in an environment 300, such as a fulfillment center floor, a warehouse environment, or another environment. To move throughout the environment 300, the foldable container 310 may be rolled using one or more wheels. For example, a manual operator may move the foldable container 310 by pushing or pulling the foldable container 310 from one location to another. The foldable container 310 may receive, carry, and transport various packages, boxes, items, products, or other objects of various types. In addition, the foldable container 310 may be configured to be lifted, carried, moved, and placed between positions within a material handling facility by robotic drive units, such as the robot 320, or other material handling equipment. In this manner, the foldable container 310 can be pushed by associates on surfaces, e.g., sort center floors, and the foldable container 310 can also be carried and transported by robotic drive units 320.

The foldable container 310 may also be moved using a robot 320, such as an autonomous robot. For example, the robot 320 may be configured to tunnel or otherwise position itself underneath the foldable container 310. The robot 320 may then lift the foldable container 310 off the ground, such that the wheels are separated from the ground, and the robot 320 may then transport the foldable container 310 from one location to another.

The foldable container 310 may maintain the ability to be moved by humans and robots, while also maintaining the ability to fold from a rectangular configuration to an L-shaped folded configuration. The foldable container 310 may include a locking bottom container platform that provides structural rigidity to enable lifting by the robot 320, while also providing the ability for the foldable container 310 to fold.

Accordingly, the foldable container 310 may be configured to be transported using both (i) the one or more, or plurality, of wheels via manual pushing, and (ii) robotic manipulation to lift and move the foldable container 310, such as by the robot 320. The foldable container 320 may be configured to allow an autonomous robot, such as the robot 320, to tunnel underneath a bottom container platform of the foldable container 310. The foldable container 310 may be configured to be lifted from the bottom container platform

by an autonomous robot or other robot 320 that is positioned at least partially or entirely underneath a center or middle portion of the foldable container 310 when the foldable container 310 is in the unfolded position (e.g., a rectangular configuration, etc.).

The wheels may include one or more freely-rotating swivel casters associated with a bottom surface of the foldable container 310. For example, the foldable container 310 may include four freely-rotating swivel casters, with one positioned at each corner of the foldable container 310. In some examples, two of the four casters may have a swivel locking feature, such as a manually or foot-operated swivel locking feature, and the other two of the four casters may have a caster rotation locking feature, such as a manually or foot-operated caster rotation locking feature.

To empty the contents of the foldable container 310, such as onto a conveyor, into another container 350 (e.g., a Gaylord, box, etc.), the foldable container 310 may be rotated at least partially upside down during a dumping process 330. The dumping process 330 may be manual or automated and may be performed using various equipment 340. During dumping, packages or other items 360 may be unloaded from the foldable container 310 and onto another surface and/or into another container.

FIGS. 4A-4B are schematic illustrations of an example container 400 having lever-based actuation of brakes and locks in accordance with one or more embodiments of the disclosure. Other embodiments may include additional or fewer components. The illustrations of FIGS. 4A-4B may not be to scale, and may not be illustrated to scale with respect to other figures. The container illustrated in FIGS. 4A-4B may be the same foldable container discussed with respect to FIGS. 1-3.

In FIG. 4A, the container 400 is depicted in both front perspective view (with the front doors) and rear perspective view (without the front doors). The foldable container 400 may include a first sidewall 410, a rear wall 420, a second sidewall 430, and front doors 440. In the rear perspective view, alternate non-mesh configurations of the first sidewall and second sidewall are depicted as alternate first sidewall 412 and alternate second sidewall 432). Some or all of the walls may be formed of panels, such as plastic panels, and may include various types of plastic or other components. For example, some portions of some or all of the walls may include plastic or other materials that are configured to facilitate sensing or other detection of contents loaded in the foldable container 400. One or more of the walls may be transparent, so as to provide visibility into the contents of the foldable container 400. The panels may be smooth and flat, so as to allow items to move smoothly along the panels. Smooth panels may be low-friction panels, and may optionally have planar contours. Some smooth panels may include perforations and/or other features. The foldable container 400 may have a greater cross-sectional area (e.g., from a top cross-sectional perspective, a side cross-sectional perspective, etc.) at or near a bottom or lower portion of the foldable container 400 than at or near a top or upper portion of the foldable container 400. This may provide a funnel-like effect when dumping items out of the foldable container 400, so as to improve item flow and reduce a likelihood for jams. The four walls of the foldable container 400 may form a rectangular configuration when the foldable container 400 is in an unfolded configuration. Accordingly, items, such as packages, may be disposed in an inside portion 450 of the foldable container 400 when the foldable container 400 is in the unfolded position. The foldable container 400 may include one or more wheels 470 that allow the foldable

container 400 to be manually pushed and/or pulled. One or more of the first sidewall 410, the rear wall 420, and/or the second sidewall 430 may include jogs or angled surfaces that facilitate a flow of items out of a top of the foldable container 400 during a dumping process.

The rear wall 420 may include one or more smooth panels along a rear of the foldable container 400, and may also include one or more lips 460 that extend from the rear panel and cover a joint between the first sidewall 410 and the rear wall 420, and/or a joint between the second sidewall 430 and the rear wall 420. The lip 460 may prevent packages or other items from getting caught or pinched between the walls of the container. The lips 460 may be integrated with the panel(s) of the rear wall 420 or may be separate components coupled to the panel(s) of the rear wall 420. The lip(s) 460 may be a first smooth panel extension disposed perpendicular to a first flat portion of the panel of the rear wall 420, where the first smooth panel extension covers at least a portion of a first joint between the first sidewall 410 and the rear wall 420. The lip(s) may include a second smooth panel extension disposed perpendicular to the first flat portion, where the second smooth panel extension covers at least a portion of the second joint.

As depicted in the rear perspective view, the rear wall 420 may include one or more structural support members 422, 426 disposed behind the panel of the rear wall 420, so as to avoid interference with packages flowing along the panel. The panel of the rear wall 420 may include an angled surface 424 or jog that reduces a cross-sectional area of the foldable container 400 at an upper side relative to a lower side of the foldable container 400.

The second sidewall 430 may include one or more ergonomic handles 434 that are disposed on an outer side of the panel(s) of the second sidewall 430. The ergonomic handles 434 may be formed so as to not extend beyond a frame of the second sidewall 430, and may be disposed relatively higher than a jog formed in the panel of the second sidewall 430. This may be possible due to the jog creating space for the ergonomic handles 434. A hook 436 or other coupling mechanism to couple foldable containers together may be coupled to the second sidewall 430. The second sidewall 430 may be configured to rotate with respect to the rear wall 420.

The foldable container 400 may include a bottom container platform that forms a bottom of the foldable container 400. The bottom container platform may be coupled to the rear wall 420 and may swing or rotate upwards and downwards relative to the rear wall 420.

The first sidewall 410 may be fixed relative to the rear wall 420 (e.g., may have a fixed position and not rotate with respect to the rear wall 420, etc.). For example, the first sidewall 410 may form an L-shaped configuration with the rear wall 420, and may remain at a substantially perpendicular angle relative to the rear wall 420 when the foldable container 400 is in a folded configuration. A fixed hinge or other mechanism may be disposed at a connection between the first sidewall 410 and the rear wall 420. In some embodiments, electronic devices, such as tracking devices, may be coupled to the foldable container 400 at or near the fixed hinge, or elsewhere on the foldable container 400. For example, some embodiments may include Bluetooth-based devices that are configured to provide location data, or data that may be used to locate or identify a particular foldable container, disposed at or near the fixed hinge. Other embodiments may include RFID devices, NFC devices, or other transceiver components that can be used to provide posi-

tioning data for a foldable container. Such data may be used to identify the location of a foldable container inside a facility.

The second sidewall 430 may be configured to swing or rotate with respect to the rear wall 420. For example, the second sidewall 430 may be configured to rotate towards the rear wall 420, so as to facilitate folding or collapsing of the foldable container 400 into the folded configuration.

The front doors 440 may be formed of one or more panels. For example, the front doors 440 may include a first panel 442 and a second panel 444 that form a first door. The first panel 442 and the second panel 444 may be coupled to the first sidewall 410. The first panel 442 and the second panel 444 may be coupled to one another or may be decoupled, such that the first panel 442 may rotate with respect to the second panel 444. The first panel 442 and the second panel 444 may be configured to rotate with respect to the first sidewall 410. For example, the first panel 442 and the second panel 444 may be configured to rotate until parallel with the first sidewall 410.

The front doors 440 may include a third panel 448 and a fourth panel 446 that form a second door. The third panel 448 and a fourth panel 446 may be coupled to the second sidewall 430. The third panel 448 and a fourth panel 446 may be coupled to one another or may be decoupled, such that the third panel 448 may rotate with respect to the fourth panel 446. The third panel 448 and a fourth panel 446 may be configured to rotate with respect to the second sidewall 430. For example, the third panel 448 and a fourth panel 446 may be configured to rotate until parallel with the second sidewall 430.

The front doors 440 may therefore include a first portion, such as a right-hand side portion with a first upper panel and a first lower panel, where the first lower panel may rotate independent of the first upper panel. The front doors 440 may include a second portion, such as a left-hand side portion with a second upper panel and a second lower panel, where the second lower panel rotates independent of both the second upper panel and the first lower panel. In some embodiments, the foldable container may include a clip configured to couple an upper panel to a corresponding lower panel, so as to prevent unintentional closing of the upper panels and/or lower panels of the front doors.

To convert from the depicted unfolded configuration to a folded configuration, the second door may be rotated parallel to the second sidewall 430, and the second sidewall 430 may be rotated to be parallel with the rear wall 420. The first door may be rotated to be parallel with the first sidewall 410. The bottom container platform may be rotated to be parallel with the rear wall 420.

The foldable container 400 may include a plurality of wheels 470 that are coupled to one or more of the walls. The foldable container 400 may be configured to be transported using (i) the plurality of wheels via manual pushing, and (ii) robotic manipulation to lift and move the foldable container.

The wheels 470 may be self-positioning swivel casters may have associated home positions that may be defined positions or orientations of the swivel casters when lifted, stationary, and/or stopped. For example, the home positions of self-positioning swivel casters associated with a container, cart, or other material handling equipment may increase or maximize space or clearance under the cart when the cart is stationary or stopped. One or more of the wheels 470 may be coupled to one or more caster lock assemblies configured to prevent swiveling of the respective wheel(s) and/or one or more brake assemblies configured to prevent rotation or rolling of the respective wheel(s), as discussed in

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more detail with respect to FIGS. 5-6. In some example embodiments, robotic drive units or other types of material handling equipment, e.g., manual, automated, or robotic equipment, may be inserted or placed under the cart and between the self-positioning swivel casters, such that increasing or maximizing space or clearance under the cart and between the self-positioning swivel casters may facilitate reliable operations and prevent collisions or interferences between robotic drive units and portions of the cart. In other example embodiments, the home positions of self-positioning swivel casters associated with a container, cart, or other material handling equipment may be positioned substantially within an outer periphery, dimensions, or footprint associated with a receptacle of the cart when the cart is stationary or stopped, such that a plurality of carts may be positioned or stored adjacent to each other with minimal gap or clearance therebetween and without causing interference between respective swivel casters associated with adjacent carts. In addition, the home positions of self-positioning swivel casters associated with a cart or other material handling equipment may enable a cart to substantially remain in position when lifted, stationary, or stopped. For example, a cart may be placed at a position on a substantially flat or horizontal floor or other surface of a material handling facility. However, due to irregularities, imperfections, protrusions, indentations, edges, slopes, angles, cracks, debris, or other surface characteristics associated with the surface, a cart that is stationary or stopped may move or drift, e.g., due to gravity. The home positions of self-positioning swivel casters may be selected or configured to prevent such movement or drift of a cart due to surface characteristics associated with a position at which the cart is placed.

The foldable container 400 may include a rod 496 configured to move vertically with respect to the bottom container platform 496. The rod 496 may be configured to move from a first position when the front doors 440 are unsecured to a second position when the front doors 440 are secured. In particular, a latch 480 may actuate the rod 496 to the second position or downwards when the latch 480 is securing the front doors 440. In the second position, the rod 496 may cause movement of a pivotable arm that blocks and unblocks a machine-readable code that may be used to identify the foldable container 400. When the machine-readable code is blocked, the foldable cart 400 may not be moved by an autonomous robot, or the autonomous robot may implement another action, such as moving at a reduced speed, generating a notification for manual intervention, generating an audible or visual alert, and so forth. In this manner, unless the front doors 440 are secured, the foldable container 400 may not be transported by an autonomous robot. The rod 496 may be a single piece assembly or a multi-piece assembly. For example, the rod 496 may have a first portion and a second portion that is discrete and separated from the first portion by a spring module that biases the rod 496 upwards or towards the first position (e.g., a multi-piece assembly formed of discrete rod segments, etc.). One or more tabs may provide hard stops for the front doors 440 when fully closed. In embodiments with multi-piece rod assemblies, a portion of the rod 496 may be coupled to the second upper panel 448 and/or the latch 480, and another portion of the rod 496 may be coupled to the second lower panel 446, where the portions may align when the front doors are in the closed position.

The container 400 may include one or more levers 490 configured to actuate caster wheel locks and/or wheel brakes of the container 400. For example, the container 400 may include a first lever configured to actuate or engage caster

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locks, a second lever configured to actuate or engage wheel brakes, and a third lever configured to disengage both the caster locks and the wheel brakes. Some embodiments may include levers on both sides of the container 400, as depicted in FIG. 4B, where the levers 490 are a first set of levers and a second set of levers 492 are disposed on an opposite side of the container 400. The levers may perform the same functions on both sides of the container 400.

In FIG. 4B, the foldable container 400 is depicted in a front view, with a door panel depicted in close-up view. The door panel may be an upper panel of the foldable container 400 or the third panel 448 of the foldable container 400 of FIG. 4A. The third panel 448 may include a frame and a hand hold or grip coupled to the frame. The grip may be a curved handle that can be used to ergonomically push and/or pull the foldable container 400. The grip may be spaced from other components of the third panel 448, such as the frame, so as to create clearance for a hand and to avoid pinch points.

The latch 480 may be coupled to the third panel 448 and may be used to secure the third panel 448 to an adjacent panel, such as the first upper panel and/or the panel from the adjacent side of the front doors 440. The latch 480 may include a palm engagement or hand hold or planar portion 482 that can be used to secure and unsecure the latch 480. The second upper panel 448 may have a tab 484 that causes the second lower panel 446 to rotate to a closed position with the second upper panel 448.

FIG. 5 is a schematic illustration of a container 500 having lever-based actuation of brakes and locks in accordance with one or more embodiments of the disclosure. Other embodiments may include additional or fewer components. The illustrations of FIG. 5 may not be to scale, and may not be illustrated to scale with respect to other figures. The container portion illustrated in FIG. 5 may be the same container discussed with respect to FIGS. 1-4B.

In FIG. 5, the container 500 may include a lever-based brake and lock assembly actuation system. In one embodiment, the container 500 may include levers connected to the brake and/or lock mechanisms on the casters via a series of cables. The cables can be routed inside or through the tube-structure of the container 500, protecting the cables against damage and inadvertent actuation. For example, the container 500 may include one or more frame portions formed of tubes or tubular structures, and the cables may be routed through the tubular structures. In some embodiments, the lever-based actuation system may have a default position of bias or spring tension with the engaged position of the lock and/or brake assembly corresponding to a latched position of increased spring tension, and the disengaged position corresponding to a released tension state. Force may be transmitted via cable tension, and in one embodiment, a lever may be a single throw lever, such that if pushing down actuates the intended mechanism, pulling up will do nothing, and instead a separate release mechanism or lever may be used. Other embodiments may be configured to have dual action levers in that moving the lever in one direction (e.g., up or down, left or right, etc.) causes actuation for engagement, and the other direction causes actuation for disengagement.

The container 500 may include, like the container discussed with respect to FIGS. 4A-4B, a first container wall, a second container wall coupled to the first container wall, the second container wall configured to rotate outwards in a first direction with respect to the first container wall, a third container wall coupled to the first container wall, and a fourth container wall configured to move from an open position to a closed position. The container 500 may include

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a bottom container platform. The container 500 may include one or more wheels, such as caster wheels. For example, the container 500 may include a first wheel 560 disposed on a first side of the container 500, such as a front side, and a second wheel 566 disposed on a second side of the container 500, such as a rear side. The container 500 may include a wheel lock assembly 520 configured to prevent swiveling of the first wheel 560. The wheel lock assembly 520 may include a first wheel lock, a first lever 590 configured to actuate the first wheel lock 520, and a first cable 574 that is coupled to the first wheel lock 520 and the first lever 590. The first lever 590 may be disposed on the third container wall, such as a lower portion of the third container wall or a sidewall of the container 500. Any number of wheel lock assemblies may be included. For example, the container 500 may include a third wheel 562 disposed on the first side of the container 500, and the container 500 may include a second wheel lock assembly that has a second wheel lock 530, where the second wheel lock is coupled to the first lever 590 via the first cable 574, such that actuation of the first lever 590 actuates both the first wheel lock 520 and the second wheel lock 530.

The container 500 may include a wheel brake assembly 540 configured to prevent rotation or rolling of the second wheel 566. The wheel brake assembly 540 may include a first wheel brake, a second lever 592 configured to actuate the first wheel brake, and a second cable 582 that is coupled to the first wheel brake and the second lever 592. The second lever 592 may be disposed on the third container wall or another location, and may be positioned adjacent to the first lever 590. Any number of wheel brake assemblies may be included. For example, the container 500 may include a fourth wheel 564 disposed on the second side of the container 500, and the container 500 may include a second wheel brake assembly that has a second wheel brake 550, where the second wheel brake 550 may be coupled to the second lever 592 via the second cable 582, such that actuation of the second lever 592 actuates both the first wheel brake 540 and the second wheel brake 550.

The container 500 may include a disengagement assembly configured to disengage the first wheel lock 520 and the first wheel brake 540. The disengagement assembly may include a third lever 594 and a third cable 576, 584 that is coupled to the first wheel lock 520, the first wheel brake 540, and the third lever 594. The third lever is disposed on the third container wall and/or otherwise adjacent to the first lever 590 and the second lever 592. Some embodiments may not include a third lever or a disengagement assembly, and instead, actuation of the first or second levers in an opposite direction may disengage the respective wheel lock or wheel brake assemblies. For example, actuation of the first lever in a first direction may cause the first wheel lock to engage, and actuation of the first lever in a second direction may cause the first wheel lock to disengage.

Levers may be included on more than one side of the container 500. For example, the wheel lock assembly and/or the container 500 may include a fourth lever disposed on the second container wall, the fourth lever coupled to the first cable 590, so as to engage the wheel locks 520, 530. A fifth lever may be disposed on the second container wall, and the fifth lever may be coupled to the second cable 582, so as to engage the wheel brakes 540, 550. The disengagement assembly and/or the container 500 may include a sixth lever disposed on the second container wall, where the sixth lever may be coupled to the third cable 576, 584, so as to disengage both the wheel locks 520, 530 and the wheel brakes 540, 550 at the same time. Any number of levers may

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be included. In some embodiments, the levers may extend from corner to corner of the container 500. For example, the first lever 590 may extend from a first corner to a second corner of the container 500.

The container 500 may have at least a partially tubular frame, where one or more of the cables may be routed through a portion of the tubular frame. For example, the container 500 may include one or more of tubular members 510, and cables may be routed therethrough. The levers may optionally be disposed on short sides of the container 500. For example, the second and third container walls (or sidewalls) of the container 500 may have a first width that is less than a second width of the first container wall (or front/rear walls) of the container 500.

In some embodiments, cables may be coupled to more than one assembly. For example, the second wheel lock 530 and the second brake 550 may be coupled via cables 578 and 580, where cable 578 may engage the second wheel lock 530 and the second brake 550, and cable 580 may disengage the second wheel lock 530 and the second brake 550. Similarly, the first wheel lock 520 and the first brake 540 may be coupled via cables 570, 572, where cable 570 may engage the first wheel lock 520 and the first wheel brake 540, and cable 572 may disengage the first wheel lock 520 and the first wheel brake 540. In this manner, a lock and brake on one side of the cart may be engaged or disengaged with a single lever if desired during maneuvering of the container 500.

FIG. 6 is a schematic illustration of various views of a container 600 having lever-based actuation of brakes and locks in accordance with one or more embodiments of the disclosure. Other embodiments may include additional or fewer components. The illustration of FIG. 6 may not be to scale, and may not be illustrated to scale with respect to other figures. The components illustrated in FIG. 6 may be used with any of the containers discussed with respect to FIGS. 1-5.

In FIG. 6, the container 600 may be a cart having a brake actuator 610 and a brake release 620 configured to engage and disengage, respectively, a brake for a wheel 630. Corresponding caster lock actuators and releases may be included as well for caster wheels. The container may include one or more levers on a sidewall 640, as discussed. For example, an engage caster locks lever, engage brakes lever, and "release all" or release both locks and brakes lever may be included. The levers may be accessible from a front 650 and/or rear side of the container 600 as well.

The container 600 may therefore be a foldable container having a rear wall, a first sidewall coupled to the rear wall, the first sidewall configured to rotate outwards in a first direction with respect to the rear wall, a second sidewall coupled to the rear wall, and a front wall that includes a first portion coupled to the first sidewall, where the first portion rotates with respect to the first sidewall, and a second portion coupled to the second sidewall, where the second portion rotates with respect to the second sidewall. The container 600 may include a bottom container platform configured to rotate from an upwards position adjacent to the rear wall to a flat position that is perpendicular to the rear wall.

The container 600 may include a set of castor wheels, such as a first castor wheel and a second castor wheel disposed on a first side of the foldable container, and a third castor wheel and a fourth castor wheel disposed on a second side of the foldable container. The container 600 may include a first castor lock configured to prevent swiveling of the first castor wheel, a second castor lock configured to prevent swiveling of the second castor wheel, a first brake

configured to prevent rotation of the third castor wheel, and a second brake configured to prevent rotation of the fourth castor wheel.

The container **600** may include a castor lock engagement assembly having a first lever disposed about a lower portion of the first sidewall, and a first cable coupled to the first lever, the first castor lock, and the second castor lock, where actuation of the first lever causes the first castor lock and the second castor lock to respectively engage the first castor wheel and the second castor wheel. The castor lock engagement assembly may include a fourth lever disposed about a lower portion of the second sidewall, the fourth lever coupled to the first cable.

The container **600** may include a brake engagement assembly having a second lever disposed about the lower portion of the first sidewall, and a second cable coupled to the second lever, the first brake, and the second brake, where actuation of the second lever causes the first brake and the second brake to respectively engage the third castor wheel and the fourth castor wheel. The brake engagement assembly may include a fifth lever disposed about the lower portion of the second sidewall, the fifth lever coupled to the second cable.

The container **600** may include a disengagement assembly having a third lever disposed about the lower portion of the first sidewall, and a third cable coupled to the third lever, the first castor lock, the second castor lock, the first brake, and the second brake, where actuation of the third lever causes the first castor lock, the second castor lock, the first brake, and the second brake to disengage. The disengagement assembly may include a sixth lever disposed about the lower portion of the second sidewall, the sixth lever coupled to the third cable.

One or more operations of the methods, process flows, or use cases of FIGS. 1-6 may have been described above as being performed by a user device, or more specifically, by one or more program module(s), applications, or the like executing on a device. It should be appreciated, however, that any of the operations of the methods, process flows, or use cases of FIGS. 1-6 may be performed, at least in part, in a distributed manner by one or more other devices, or more specifically, by one or more program module(s), applications, or the like executing on such devices. In addition, it should be appreciated that processing performed in response to the execution of computer-executable instructions provided as part of an application, program module, or the like may be interchangeably described herein as being performed by the application or the program module itself or by a device on which the application, program module, or the like is executing. While the operations of the methods, process flows, or use cases of FIGS. 1-6 may be described in the context of the illustrative devices, it should be appreciated that such operations may be implemented in connection with numerous other device configurations.

The operations described and depicted in the illustrative methods, process flows, and use cases of FIGS. 1-6 may be carried out or performed in any suitable order, such as the depicted orders, as desired in various example embodiments of the disclosure. Additionally, in certain example embodiments, at least a portion of the operations may be carried out in parallel. Furthermore, in certain example embodiments, less, more, or different operations than those depicted in FIGS. 1-6 may be performed.

Although specific embodiments of the disclosure have been described, one of ordinary skill in the art will recognize that numerous other modifications and alternative embodiments are within the scope of the disclosure. For example,

any of the functionality and/or processing capabilities described with respect to a particular device or component may be performed by any other device or component. Further, while various illustrative implementations and architectures have been described in accordance with embodiments of the disclosure, one of ordinary skill in the art will appreciate that numerous other modifications to the illustrative implementations and architectures described herein are also within the scope of this disclosure.

Certain aspects of the disclosure are described above with reference to block and flow diagrams of systems, methods, apparatuses, and/or computer program products according to example embodiments. It will be understood that one or more blocks of the block diagrams and flow diagrams, and combinations of blocks in the block diagrams and the flow diagrams, respectively, may be implemented by the execution of computer-executable program instructions. Likewise, some blocks of the block diagrams and flow diagrams may not necessarily need to be performed in the order presented, or may not necessarily need to be performed at all, according to some embodiments. Further, additional components and/or operations beyond those depicted in blocks of the block and/or flow diagrams may be present in certain embodiments.

Accordingly, blocks of the block diagrams and flow diagrams support combinations of means for performing the specified functions, combinations of elements or steps for performing the specified functions, and program instruction means for performing the specified functions. It will also be understood that each block of the block diagrams and flow diagrams, and combinations of blocks in the block diagrams and flow diagrams, may be implemented by special-purpose, hardware-based computer systems that perform the specified functions, elements or steps, or combinations of special-purpose hardware and computer instructions.

Illustrative Computer Architecture

FIG. 7 is a schematic block diagram of one or more illustrative computer system(s) **700** in accordance with one or more example embodiments of the disclosure. The computer system(s) **700** may include any suitable computing device including, but not limited to, a server system, a voice interaction device, a mobile device such as a smartphone, a tablet, an e-reader, a wearable device, or the like; a desktop computer; a laptop computer; a content streaming device; or the like. The computer system(s) **700** may correspond to an illustrative device configuration for a computer system used in conjunction with any one of the robotic system(s) of FIGS. 1-6, such as robotic manipulators and/or autonomous robotic vehicles.

The computer system(s) **700** may be configured to communicate with one or more servers, user devices, or the like. The computer system(s) **700** may be configured to cause the robotic system(s) to deposit containers into one or more pods, retrieve containers, transport containers, dump containers, and so forth.

The computer system(s) **700** may be configured to communicate via one or more networks. Such network(s) may include, but are not limited to, any one or more different types of communications networks such as, for example, cable networks, public networks (e.g., the Internet), private networks (e.g., frame-relay networks), wireless networks, cellular networks, telephone networks (e.g., a public switched telephone network), or any other suitable private or public packet-switched or circuit-switched networks. Further, such network(s) may have any suitable communication range associated therewith and may include, for example, global networks (e.g., the Internet), metropolitan area net-

works (MANs), wide area networks (WANs), local area networks (LANs), or personal area networks (PANs). In addition, such network(s) may include communication links and associated networking devices (e.g., link-layer switches, routers, etc.) for transmitting network traffic over any suitable type of medium including, but not limited to, coaxial cable, twisted-pair wire (e.g., twisted-pair copper wire), optical fiber, a hybrid fiber-coaxial (HFC) medium, a microwave medium, a radio frequency communication medium, a satellite communication medium, or any combination thereof.

In an illustrative configuration, the computer system(s) **700** may include one or more processors (processor(s)) **702**, one or more memory devices **704** (also referred to herein as memory **704**), one or more input/output (I/O) interface(s) **706**, one or more network interface(s) **708**, one or more sensor(s) or sensor interface(s) **710**, one or more transceiver(s) **712**, one or more optional display(s) **714**, one or more optional microphone(s) **716**, and data storage **720**. The computer system(s) **700** may further include one or more bus(es) **718** that functionally couple various components of the computer system(s) **700**. The computer system(s) **700** may further include one or more antenna(s) **730** that may include, without limitation, a cellular antenna for transmitting or receiving signals to/from a cellular network infrastructure, an antenna for transmitting or receiving Wi-Fi signals to/from an access point (AP), a Global Navigation Satellite System (GNSS) antenna for receiving GNSS signals from a GNSS satellite, a Bluetooth antenna for transmitting or receiving Bluetooth signals, a Near Field Communication (NFC) antenna for transmitting or receiving NFC signals, and so forth. These various components will be described in more detail hereinafter.

The bus(es) **718** may include at least one of a system bus, a memory bus, an address bus, or a message bus, and may permit the exchange of information (e.g., data (including computer-executable code), signaling, etc.) between various components of the computer system(s) **700**. The bus(es) **718** may include, without limitation, a memory bus or a memory controller, a peripheral bus, an accelerated graphics port, and so forth. The bus(es) **718** may be associated with any suitable bus architecture including, without limitation, an Industry Standard Architecture (ISA), a Micro Channel Architecture (MCA), an Enhanced ISA (EISA), a Video Electronics Standards Association (VESA) architecture, an Accelerated Graphics Port (AGP) architecture, a Peripheral Component Interconnect (PCI) architecture, a PCI-Express architecture, a Personal Computer Memory Card International Association (PCMCIA) architecture, a Universal Serial Bus (USB) architecture, and so forth.

The memory **704** of the computer system(s) **700** may include volatile memory (memory that maintains its state when supplied with power) such as random access memory (RAM) and/or non-volatile memory (memory that maintains its state even when not supplied with power) such as read-only memory (ROM), flash memory, ferroelectric RAM (FRAM), and so forth. Persistent data storage, as that term is used herein, may include non-volatile memory. In certain example embodiments, volatile memory may enable faster read/write access than non-volatile memory. However, in certain other example embodiments, certain types of non-volatile memory (e.g., FRAM) may enable faster read/write access than certain types of volatile memory.

In various implementations, the memory **704** may include multiple different types of memory such as various types of static random access memory (SRAM), various types of dynamic random access memory (DRAM), various types of

unalterable ROM, and/or writeable variants of ROM such as electrically erasable programmable read-only memory (EEPROM), flash memory, and so forth. The memory **704** may include main memory as well as various forms of cache memory such as instruction cache(s), data cache(s), translation lookaside buffer(s) (TLBs), and so forth. Further, cache memory such as a data cache may be a multi-level cache organized as a hierarchy of one or more cache levels (L1, L2, etc.).

The data storage **720** may include removable storage and/or non-removable storage including, but not limited to, magnetic storage, optical disk storage, and/or tape storage. The data storage **720** may provide non-volatile storage of computer-executable instructions and other data. The memory **704** and the data storage **720**, removable and/or non-removable, are examples of computer-readable storage media (CRSM) as that term is used herein.

The data storage **720** may store computer-executable code, instructions, or the like that may be loadable into the memory **704** and executable by the processor(s) **702** to cause the processor(s) **702** to perform or initiate various operations. The data storage **720** may additionally store data that may be copied to the memory **704** for use by the processor(s) **702** during the execution of the computer-executable instructions. Moreover, output data generated as a result of execution of the computer-executable instructions by the processor(s) **702** may be stored initially in the memory **704**, and may ultimately be copied to the data storage **720** for non-volatile storage.

More specifically, the data storage **720** may store one or more operating systems (O/S) **722**; one or more database management systems (DBMS) **724**; and one or more program module(s), applications, engines, computer-executable code, scripts, or the like. Some or all of these module(s) may be sub-module(s). Any of the components depicted as being stored in the data storage **720** may include any combination of software, firmware, and/or hardware. The software and/or firmware may include computer-executable code, instructions, or the like that may be loaded into the memory **704** for execution by one or more of the processor(s) **702**. Any of the components depicted as being stored in the data storage **720** may support functionality described in reference to corresponding components named earlier in this disclosure.

The data storage **720** may further store various types of data utilized by the components of the computer system(s) **700**. Any data stored in the data storage **720** may be loaded into the memory **704** for use by the processor(s) **702** in executing computer-executable code. In addition, any data depicted as being stored in the data storage **720** may potentially be stored in one or more datastore(s) and may be accessed via the DBMS **724** and loaded in the memory **704** for use by the processor(s) **702** in executing computer-executable code. The datastore(s) may include, but are not limited to, databases (e.g., relational, object-oriented, etc.), file systems, flat files, distributed datastores in which data is stored on more than one node of a computer network, peer-to-peer network datastores, or the like.

The processor(s) **702** may be configured to access the memory **704** and execute the computer-executable instructions loaded therein. For example, the processor(s) **702** may be configured to execute the computer-executable instructions of the various program module(s), applications, engines, or the like of the computer system(s) **700** to cause or facilitate various operations to be performed in accordance with one or more embodiments of the disclosure. The processor(s) **702** may include any suitable processing unit capable of accepting data as input, processing the input data

in accordance with stored computer-executable instructions, and generating output data. The processor(s) 702 may include any type of suitable processing unit including, but not limited to, a central processing unit, a microprocessor, a Reduced Instruction Set Computer (RISC) microprocessor, a Complex Instruction Set Computer (CISC) microprocessor, a microcontroller, an Application Specific Integrated Circuit (ASIC), a Field-Programmable Gate Array (FPGA), a System-on-a-Chip (SoC), a digital signal processor (DSP), and so forth. Further, the processor(s) 702 may have any suitable microarchitecture design that includes any number of constituent components such as, for example, registers, multiplexers, arithmetic logic units, cache controllers for controlling read/write operations to cache memory, branch predictors, or the like. The microarchitecture design of the processor(s) 702 may be capable of supporting any of a variety of instruction sets.

Referring now to other illustrative components depicted as being stored in the data storage 720, the O/S 722 may be loaded from the data storage 720 into the memory 704 and may provide an interface between other application software executing on the computer system(s) 700 and the hardware resources of the computer system(s) 700. More specifically, the O/S 722 may include a set of computer-executable instructions for managing the hardware resources of the computer system(s) 700 and for providing common services to other application programs (e.g., managing memory allocation among various application programs). In certain example embodiments, the O/S 722 may control execution of the other program module(s). The O/S 722 may include any operating system now known or which may be developed in the future including, but not limited to, any server operating system, any mainframe operating system, or any other proprietary or non-proprietary operating system.

The DBMS 724 may be loaded into the memory 704 and may support functionality for accessing, retrieving, storing, and/or manipulating data stored in the memory 704 and/or data stored in the data storage 720. The DBMS 724 may use any of a variety of database models (e.g., relational model, object model, etc.) and may support any of a variety of query languages. The DBMS 724 may access data represented in one or more data schemas and stored in any suitable data repository including, but not limited to, databases (e.g., relational, object-oriented, etc.), file systems, flat files, distributed datastores in which data is stored on more than one node of a computer network, peer-to-peer network datastores, or the like. In those example embodiments in which the computer system(s) 700 is a mobile device, the DBMS 724 may be any suitable lightweight DBMS optimized for performance on a mobile device.

Referring now to other illustrative components of the computer system(s) 700, the input/output (I/O) interface(s) 706 may facilitate the receipt of input information by the computer system(s) 700 from one or more I/O devices as well as the output of information from the computer system(s) 700 to the one or more I/O devices. The I/O devices may include any of a variety of components such as a display or display screen having a touch surface or touchscreen; an audio output device for producing sound, such as a speaker; an audio capture device, such as a microphone; an image and/or video capture device, such as a camera; a haptic unit; and so forth. Any of these components may be integrated into the computer system(s) 700 or may be separate. The I/O devices may further include, for example, any number of peripheral devices such as data storage devices, printing devices, and so forth.

The I/O interface(s) 706 may also include an interface for an external peripheral device connection such as universal serial bus (USB), FireWire, Thunderbolt, Ethernet port or other connection protocol that may connect to one or more networks. The I/O interface(s) 706 may also include a connection to one or more of the antenna(s) 730 to connect to one or more networks via a wireless local area network (WLAN) (such as Wi-Fi) radio, Bluetooth, ZigBee, and/or a wireless network radio, such as a radio capable of communication with a wireless communication network such as a Long Term Evolution (LTE) network, WiMAX network, 3G network, a ZigBee network, etc.

The computer system(s) 700 may further include one or more network interface(s) 708 via which the computer system(s) 700 may communicate with any of a variety of other systems, platforms, networks, devices, and so forth. The network interface(s) 708 may enable communication, for example, with one or more wireless routers, one or more host servers, one or more web servers, and the like via one or more networks.

The antenna(s) 730 may include any suitable type of antenna depending, for example, on the communications protocols used to transmit or receive signals via the antenna(s) 730. Non-limiting examples of suitable antennas may include directional antennas, non-directional antennas, dipole antennas, folded dipole antennas, patch antennas, multiple-input multiple-output (MIMO) antennas, or the like. The antenna(s) 730 may be communicatively coupled to one or more transceivers 712 or radio components to which or from which signals may be transmitted or received.

As previously described, the antenna(s) 730 may include a cellular antenna configured to transmit or receive signals in accordance with established standards and protocols, such as Global System for Mobile Communications (GSM), 3G standards (e.g., Universal Mobile Telecommunications System (UMTS), Wideband Code Division Multiple Access (W-CDMA), CDMA2000, etc.), 4G standards (e.g., Long-Term Evolution (LTE), WiMax, etc.), direct satellite communications, or the like.

The antenna(s) 730 may additionally, or alternatively, include a Wi-Fi antenna configured to transmit or receive signals in accordance with established standards and protocols, such as the IEEE 802.11 family of standards, including via 2.4 GHz channels (e.g., 802.11b, 802.11g, 802.11n), 5 GHz channels (e.g., 802.11n, 802.11ac), or 60 GHz channels (e.g., 802.11ad). In alternative example embodiments, the antenna(s) 730 may be configured to transmit or receive radio frequency signals within any suitable frequency range forming part of the unlicensed portion of the radio spectrum.

The antenna(s) 730 may additionally, or alternatively, include a GNSS antenna configured to receive GNSS signals from three or more GNSS satellites carrying time-position information to triangulate a position therefrom. Such a GNSS antenna may be configured to receive GNSS signals from any current or planned GNSS such as, for example, the Global Positioning System (GPS), the GLONASS System, the Compass Navigation System, the Galileo System, or the Indian Regional Navigational System.

The transceiver(s) 712 may include any suitable radio component(s) for—in cooperation with the antenna(s) 730—transmitting or receiving radio frequency (RF) signals in the bandwidth and/or channels corresponding to the communications protocols utilized by the computer system(s) 700 to communicate with other devices. The transceiver(s) 712 may include hardware, software, and/or firmware for modulating, transmitting, or receiving—potentially in cooperation with any of antenna(s) 730—commu-

nications signals according to any of the communications protocols discussed above including, but not limited to, one or more Wi-Fi and/or Wi-Fi direct protocols, as standardized by the IEEE 802.11 standards, one or more non-Wi-Fi protocols, or one or more cellular communications protocols or standards. The transceiver(s) **712** may further include hardware, firmware, or software for receiving GNSS signals. The transceiver(s) **712** may include any known receiver and baseband suitable for communicating via the communications protocols utilized by the computer system(s) **700**. The transceiver(s) **712** may further include a low noise amplifier (LNA), additional signal amplifiers, an analog-to-digital (A/D) converter, one or more buffers, a digital baseband, or the like.

The sensor(s)/sensor interface(s) **710** may include or may be capable of interfacing with any suitable type of sensing device such as, for example, inertial sensors, force sensors, thermal sensors, photocells, and so forth. Example types of inertial sensors may include accelerometers (e.g., MEMS-based accelerometers), gyroscopes, and so forth.

The optional display(s) **714** may be configured to output light and/or render content. The optional speaker(s)/microphone(s) **716** may be any device configured to receive analog sound input or voice data.

It should be appreciated that the program module(s), applications, computer-executable instructions, code, or the like depicted in FIG. 7 as being stored in the data storage **720** are merely illustrative and not exhaustive and that processing described as being supported by any particular module may alternatively be distributed across multiple module(s) or performed by a different module. In addition, various program module(s), script(s), plug-in(s), Application Programming Interface(s) (API(s)), or any other suitable computer-executable code hosted locally on the computer system(s) **700**, and/or hosted on other computing device(s) accessible via one or more networks, may be provided to support functionality provided by the program module(s), applications, or computer-executable code depicted in FIG. 7 and/or additional or alternate functionality. Further, functionality may be modularized differently such that processing described as being supported collectively by the collection of program module(s) depicted in FIG. 7 may be performed by a fewer or greater number of module(s), or functionality described as being supported by any particular module may be supported, at least in part, by another module. In addition, program module(s) that support the functionality described herein may form part of one or more applications executable across any number of systems or devices in accordance with any suitable computing model such as, for example, a client-server model, a peer-to-peer model, and so forth. In addition, any of the functionality described as being supported by any of the program module(s) depicted in FIG. 7 may be implemented, at least partially, in hardware and/or firmware across any number of devices.

It should further be appreciated that the computer system(s) **700** may include alternate and/or additional hardware, software, or firmware components beyond those described or depicted without departing from the scope of the disclosure. More particularly, it should be appreciated that software, firmware, or hardware components depicted as forming part of the computer system(s) **700** are merely illustrative and that some components may not be present or additional components may be provided in various embodiments. While various illustrative program module(s) have been depicted and described as software module(s) stored in the data storage **720**, it should be appreciated that function-

ality described as being supported by the program module(s) may be enabled by any combination of hardware, software, and/or firmware. It should further be appreciated that each of the above-mentioned module(s) may, in various embodiments, represent a logical partitioning of supported functionality. This logical partitioning is depicted for ease of explanation of the functionality and may not be representative of the structure of software, hardware, and/or firmware for implementing the functionality. Accordingly, it should be appreciated that functionality described as being provided by a particular module may, in various embodiments, be provided at least in part by one or more other module(s). Further, one or more depicted module(s) may not be present in certain embodiments, while in other embodiments, additional module(s) not depicted may be present and may support at least a portion of the described functionality and/or additional functionality. Moreover, while certain module(s) may be depicted and described as sub-module(s) of another module, in certain embodiments, such module(s) may be provided as independent module(s) or as sub-module(s) of other module(s).

One or more operations of the methods, process flows, and use cases of FIGS. 1-6 may be performed by a device having the illustrative configuration depicted in FIG. 7, or more specifically, by one or more engines, program module(s), applications, or the like executable on such a device. It should be appreciated, however, that such operations may be implemented in connection with numerous other device configurations.

The operations described and depicted in the illustrative methods and process flows of any of FIGS. 1-6 may be carried out or performed in any suitable order as desired in various example embodiments of the disclosure. Additionally, in certain example embodiments, at least a portion of the operations may be carried out in parallel. Furthermore, in certain example embodiments, less, more, or different operations than those depicted in FIGS. 1-6 may be performed.

Although specific embodiments of the disclosure have been described, one of ordinary skill in the art will recognize that numerous other modifications and alternative embodiments are within the scope of the disclosure. For example, any of the functionality and/or processing capabilities described with respect to a particular device or component may be performed by any other device or component. Further, while various illustrative implementations and architectures have been described in accordance with embodiments of the disclosure, one of ordinary skill in the art will appreciate that numerous other modifications to the illustrative implementations and architectures described herein are also within the scope of this disclosure.

Certain aspects of the disclosure are described above with reference to block and flow diagrams of systems, methods, apparatuses, and/or computer program products according to example embodiments. It will be understood that one or more blocks of the block diagrams and flow diagrams, and combinations of blocks in the block diagrams and the flow diagrams, respectively, may be implemented by execution of computer-executable program instructions. Likewise, some blocks of the block diagrams and flow diagrams may not necessarily need to be performed in the order presented, or may not necessarily need to be performed at all, according to some embodiments. Further, additional components and/or operations beyond those depicted in blocks of the block and/or flow diagrams may be present in certain embodiments.

Accordingly, blocks of the block diagrams and flow diagrams support combinations of means for performing the

specified functions, combinations of elements or steps for performing the specified functions, and program instruction means for performing the specified functions. It will also be understood that each block of the block diagrams and flow diagrams, and combinations of blocks in the block diagrams and flow diagrams, may be implemented by special-purpose, hardware-based computer systems that perform the specified functions, elements or steps, or combinations of special-purpose hardware and computer instructions.

Program module(s), applications, or the like disclosed herein may include one or more software components including, for example, software objects, methods, data structures, or the like. Each such software component may include computer-executable instructions that, responsive to execution, cause at least a portion of the functionality described herein (e.g., one or more operations of the illustrative methods described herein) to be performed.

A software component may be coded in any of a variety of programming languages. An illustrative programming language may be a lower-level programming language such as an assembly language associated with a particular hardware architecture and/or operating system platform. A software component comprising assembly language instructions may require conversion into executable machine code by an assembler prior to execution by the hardware architecture and/or platform.

Another example programming language may be a higher-level programming language that may be portable across multiple architectures. A software component comprising higher-level programming language instructions may require conversion to an intermediate representation by an interpreter or a compiler prior to execution.

Other examples of programming languages include, but are not limited to, a macro language, a shell or command language, a job control language, a script language, a database query or search language, or a report writing language. In one or more example embodiments, a software component comprising instructions in one of the foregoing examples of programming languages may be executed directly by an operating system or other software component without having to be first transformed into another form.

A software component may be stored as a file or other data storage construct. Software components of a similar type or functionally related may be stored together such as, for example, in a particular directory, folder, or library. Software components may be static (e.g., pre-established or fixed) or dynamic (e.g., created or modified at the time of execution).

Software components may invoke or be invoked by other software components through any of a wide variety of mechanisms. Invoked or invoking software components may comprise other custom-developed application software, operating system functionality (e.g., device drivers, data storage (e.g., file management) routines, other common routines and services, etc.), or third-party software components (e.g., middleware, encryption, or other security software, database management software, file transfer or other network communication software, mathematical or statistical software, image processing software, and format translation software).

Software components associated with a particular solution or system may reside and be executed on a single platform or may be distributed across multiple platforms. The multiple platforms may be associated with more than one hardware vendor, underlying chip technology, or operating system. Furthermore, software components associated with a particular solution or system may be initially written

in one or more programming languages, but may invoke software components written in another programming language.

Computer-executable program instructions may be loaded onto a special-purpose computer or other particular machine, a processor, or other programmable data processing apparatus to produce a particular machine, such that execution of the instructions on the computer, processor, or other programmable data processing apparatus causes one or more functions or operations specified in the flow diagrams to be performed. These computer program instructions may also be stored in a computer-readable storage medium (CRSM) that upon execution may direct a computer or other programmable data processing apparatus to function in a particular manner, such that the instructions stored in the computer-readable storage medium produce an article of manufacture including instruction means that implement one or more functions or operations specified in the flow diagrams. The computer program instructions may also be loaded onto a computer or other programmable data processing apparatus to cause a series of operational elements or steps to be performed on the computer or other programmable apparatus to produce a computer-implemented process.

Additional types of CRSM that may be present in any of the devices described herein may include, but are not limited to, programmable random access memory (PRAM), SRAM, DRAM, RAM, ROM, electrically erasable programmable read-only memory (EEPROM), flash memory or other memory technology, compact disc read-only memory (CD-ROM), digital versatile disc (DVD) or other optical storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices, or any other medium which can be used to store the information and which can be accessed. Combinations of any of the above are also included within the scope of CRSM. Alternatively, computer-readable communication media (CRCM) may include computer-readable instructions, program module(s), or other data transmitted within a data signal, such as a carrier wave, or other transmission. However, as used herein, CRSM does not include CRCM.

Although embodiments have been described in language specific to structural features and/or methodological acts, it is to be understood that the disclosure is not necessarily limited to the specific features or acts described. Rather, the specific features and acts are disclosed as illustrative forms of implementing the embodiments. Conditional language, such as, among others, “can,” “could,” “might,” or “may,” unless specifically stated otherwise, or otherwise understood within the context as used, is generally intended to convey that certain embodiments could include, while other embodiments do not include, certain features, elements, and/or steps. Thus, such conditional language is not generally intended to imply that features, elements, and/or steps are in any way required for one or more embodiments or that one or more embodiments necessarily include logic for deciding, with or without user input or prompting, whether these features, elements, and/or steps are included or are to be performed in any particular embodiment.

That which is claimed is:

1. A foldable container comprising:

a rear wall;

a first sidewall coupled to the rear wall, the first sidewall configured to rotate outwards in a first direction with respect to the rear wall;

a second sidewall coupled to the rear wall;

a front wall comprising:

a first portion coupled to the first sidewall, wherein the first portion rotates with respect to the first sidewall; and

a second portion coupled to the second sidewall, wherein the second portion rotates with respect to the second sidewall;

a bottom container platform configured to rotate from an upwards position adjacent to the rear wall to a flat position that is perpendicular to the rear wall;

a set of castor wheels comprising:

a first castor wheel and a second castor wheel disposed on a first side of the foldable container; and

a third castor wheel and a fourth castor wheel disposed on a second side of the foldable container;

a first castor lock configured to prevent swiveling of the first castor wheel;

a second castor lock configured to prevent swiveling of the second castor wheel;

a first brake configured to prevent rotation of the third castor wheel;

a second brake configured to prevent rotation of the fourth castor wheel;

a castor lock engagement assembly comprising:

a first lever disposed about a lower portion of the first sidewall; and

a first cable coupled to the first lever, the first castor lock, and the second castor lock, wherein actuation of the first lever causes the first castor lock and the second castor lock to respectively engage the first castor wheel and the second castor wheel;

a brake engagement assembly comprising:

a second lever disposed about the lower portion of the first sidewall; and

a second cable coupled to the second lever, the first brake, and the second brake, wherein actuation of the second lever causes the first brake and the second brake to respectively engage the third castor wheel and the fourth castor wheel; and

a disengagement assembly comprising:

a third lever disposed about the lower portion of the first sidewall; and

a third cable coupled to the third lever, the first castor lock, the second castor lock, the first brake, and the second brake, wherein actuation of the third lever causes the first castor lock, the second castor lock, the first brake, and the second brake to disengage.

2. The foldable container of claim 1, wherein:

the castor lock engagement assembly further comprises a fourth lever disposed about a lower portion of the second sidewall, the fourth lever coupled to the first cable;

the brake engagement assembly further comprises a fifth lever disposed about the lower portion of the second sidewall, the fifth lever coupled to the second cable; and

the disengagement assembly further comprises a sixth lever disposed about the lower portion of the second sidewall, the sixth lever coupled to the third cable.

3. The foldable container of claim 1, wherein the first sidewall comprises a tubular frame, and wherein the first cable, the second cable, and the third cable are routed through a portion of the tubular frame.

4. The foldable container of claim 1, wherein the foldable container is configured to be lifted from the bottom container platform by an autonomous robot that is positioned underneath the foldable container when the foldable container is in an unfolded position; and

wherein the foldable container is configured to be transported using (i) the plurality of wheels via manual pushing, and (ii) robotic manipulation to lift and move the foldable container.

5. A container comprising:

a first container wall;

a second container wall coupled to the first container wall, the second container wall configured to rotate outwards in a first direction with respect to the first container wall;

a third container wall coupled to the first container wall; a fourth container wall configured to move from an open position to a closed position;

a bottom container platform;

a first wheel disposed on a first side of the container;

a second wheel disposed on a second side of the container;

a wheel lock assembly configured to prevent swiveling of the first wheel, the wheel lock assembly comprising a first wheel lock, a first lever configured to actuate the first wheel lock, and a first cable that is coupled to the first wheel lock and the first lever, wherein the first lever is disposed on the third container wall; and

a wheel brake assembly configured to prevent rotation of the second wheel, the wheel brake assembly comprising a first wheel brake, a second lever configured to actuate the first wheel brake, and a second cable that is coupled to the first wheel brake and the second lever, wherein the second lever is disposed on the third container wall.

6. The container of claim 5, further comprising:

a disengagement assembly configured to disengage the first wheel lock and the first wheel brake, the disengagement assembly comprising a third lever and a third cable that is coupled to the first wheel lock, the first wheel brake, and the third lever, wherein the third lever is disposed on the third container wall.

7. The container of claim 6, wherein:

the wheel lock assembly further comprises a fourth lever disposed on the second container wall, the fourth lever coupled to the first cable;

the wheel brake assembly further comprises a fifth lever disposed on the second container wall, the fifth lever coupled to the second cable; and

the disengagement assembly further comprises a sixth lever disposed on the second container wall, the sixth lever coupled to the third cable.

8. The container of claim 7, wherein the first lever extends from a first corner to a second corner of the container, and the fourth lever extends from a third corner to a fourth corner of the container.

9. The container of claim 5, further comprising:

a third wheel disposed on the first side of the container; and

a fourth wheel disposed on the second side of the container;

wherein the wheel lock assembly further comprises a second wheel lock disposed adjacent to the third wheel, the second wheel lock coupled to the first cable, and the

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wheel brake assembly further comprises a second wheel brake disposed adjacent to the fourth wheel, the second wheel brake coupled to the second cable.

10. The container of claim 5, wherein actuation of the first lever in a first direction causes the first wheel lock to engage, and actuation of the first lever in a second direction causes the first wheel lock to disengage.

11. The container of claim 5, wherein the first container wall comprises a tubular frame, and wherein the first cable and the second cable are routed through a portion of the tubular frame.

12. The container of claim 5, wherein the second container wall has a first width that is less than a second width of the first container wall.

13. The container of claim 5, wherein the fourth container wall comprises:

a first portion coupled to the second container wall; and
a second portion coupled to the third container wall, wherein the first portion and the second portion are coplanar when the fourth container wall is in the closed position.

14. The container of claim 5, wherein the bottom container platform is configured to rotate outwards in a second direction with respect to the first container wall, the bottom container platform comprising an automatic locking mechanism configured to engage with the second container wall when the container is in an unfolded position.

15. The container of claim 5, wherein the container is configured to be transported using (i) a plurality of wheels via manual pushing, and (ii) robotic manipulation to lift and move the container.

16. A container comprising:

a first container wall;
a second container wall coupled to the first container wall, the second container wall configured to rotate outwards in a first direction with respect to the first container wall;

a third container wall coupled to the first container wall;
a fourth container wall configured to move from an open position to a closed position, the fourth container wall comprising:

a first portion coupled to the second container wall; and
a second portion coupled to the third container wall, wherein the first portion and the second portion are coplanar when the fourth container wall is in the closed position;

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a first wheel disposed on a first side of the container;
a second wheel disposed on a second side of the container;
a wheel lock assembly configured to prevent swiveling of

the first wheel, the wheel lock assembly comprising a first wheel lock, a first lever configured to actuate the first wheel lock, and a first cable that is coupled to the first wheel lock and the first lever, wherein the first lever is disposed on the third container wall; and

a wheel brake assembly configured to prevent rotation of the second wheel, the wheel brake assembly comprising a first wheel brake, a second lever configured to actuate the first wheel brake, and a second cable that is coupled to the first wheel brake and the second lever, wherein the second lever is disposed on the third container wall.

17. The container of claim 16, further comprising:

a disengagement assembly configured to disengage the first wheel lock and the first wheel brake, the disengagement assembly comprising a third lever and a third cable that is coupled to the first wheel lock, the first wheel brake, and the third lever, wherein the third lever is disposed on the third container wall.

18. The container of claim 16, further comprising:

a third wheel disposed on the first side of the container; and

a fourth wheel disposed on the second side of the container;

wherein the wheel lock assembly further comprises a second wheel lock disposed adjacent to the third wheel, the second wheel lock coupled to the first cable, and the wheel brake assembly further comprises a second wheel brake disposed adjacent to the fourth wheel, the second wheel brake coupled to the second cable.

19. The container of claim 16, wherein actuation of the first lever in a first direction causes the first wheel lock to engage, and actuation of the first lever in a second direction causes the first wheel lock to disengage.

20. The container of claim 16, wherein the first container wall comprises a tubular frame, and wherein the first cable and the second cable are routed through a portion of the tubular frame.

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